

Chapter 9

The Fundamental Group

One of the basic problems of topology is to determine whether two given topological spaces are homeomorphic or not. There is no method for solving this problem in general, but techniques do exist that apply in particular cases.

Showing that two spaces *are* homeomorphic is a matter of constructing a continuous mapping from one to the other having a continuous inverse, and constructing continuous functions is a problem that we have developed techniques to handle.

Showing that two spaces are *not* homeomorphic is a different matter. For that, one must show that a continuous function with continuous inverse does *not* exist. If one can find some topological property that holds for one space but not for the other, then the problem is solved—the spaces cannot be homeomorphic. The closed interval $[0, 1]$ cannot be homeomorphic to the open interval $(0, 1)$, for instance, because the first space is compact and the second one is not. And the real line $\mathbb{R}$ cannot be homeomorphic to the “long line” L , because $\mathbb{R}$ has a countable basis and L does not. Nor can the real line $\mathbb{R}$ be homeomorphic to the plane $\mathbb{R}^2$; deleting a point from $\mathbb{R}^2$ leaves a connected space remaining, and deleting a point from $\mathbb{R}$ does not.

But the topological properties we have studied up to now do not carry us very far in solving the problem. For instance, how does one show that the plane $\mathbb{R}^2$ is not homeomorphic to three-dimensional space $\mathbb{R}^3$? As one goes down the list of topological properties—compactness, connectedness, local connectedness, metrizable, and so on—one can find no topological property that distinguishes between them. As another example, consider such surfaces as the 2-sphere S^2 , the torus T (surface of a

doughnut), and the double torus $T\#T$ (surface of a two-holed doughnut). None of the topological properties we have studied up to now will distinguish between them.

So we must introduce new properties and new techniques. One of the most natural such properties is that of *simple connectedness*. You probably have studied this notion already, when you studied line integrals in the plane. Roughly speaking, one says that a space X is simply connected if every closed curve in X can be shrunk to a point in X . (We shall make this more precise later.) The property of simple connectedness, it turns out, will distinguish between $\mathbb{R}^2$ and $\mathbb{R}^3$, deleting a point from $\mathbb{R}^3$ leaves a simply connected space remaining, but deleting a point from $\mathbb{R}^2$ does not. It will also distinguish between S^2 (which is simply connected) and the torus T (which is not). But it will not distinguish between T and $T\#T$; neither of them is simply connected.

There is an idea more general than the idea of simple connectedness, an idea that includes simple connectedness as a special case. It involves a certain *group* that is called the *fundamental group* of the space. Two spaces that are homeomorphic have fundamental groups that are isomorphic. And the condition of simple connectedness is just the condition that the fundamental group of X is the trivial (one-element) group. Thus, the proof that S^2 and T are not homeomorphic can be rephrased by saying that the fundamental group of S^2 is trivial and the fundamental group of T is not. The fundamental group will distinguish between more spaces than the condition of simple connectedness will. It can be used, for example, to show that T and $T\#T$ are not homeomorphic; it turns out that T has an abelian fundamental group and $T\#T$ does not.

In this chapter, we define the fundamental group and study its properties. Then we apply it to a number of problems, including the problem of showing that various spaces, such as those already mentioned, are not homeomorphic.

Other applications include theorems about fixed points and antipode-preserving maps of the sphere, as well as the well-known *fundamental theorem of algebra*, which says that every polynomial equation with real or complex coefficients has a root. Finally, there is the famous *Jordan curve theorem*, which we shall study in the next chapter; it states that every simple closed curve C in the plane separates the plane into two components, of which C is the common boundary.

Throughout, we assume familiarity with the quotient topology (§22) and local connectedness (§25).

§51 Homotopy of Paths

Before defining the fundamental group of a space X , we shall consider paths on X and an equivalence relation called *path homotopy* between them. And we shall define a certain operation on the collection of the equivalence classes that makes it into what is called in algebra a *groupoid*.

Definition. If f and f' are continuous maps of the space X into the space Y , we say that f is **homotopic** to f' if there is a continuous map $F : X \times I \rightarrow Y$ such that

$$F(x, 0) = f(x) \quad \text{and} \quad F(x, 1) = f'(x)$$

for each x . (Here $I = [0, 1]$.) The map F is called a **homotopy** between f and f' . If f is homotopic to f' , we write $f \simeq f'$. If $f \simeq f'$ and f' is a constant map, we say that f is **nulhomotopic**.

We think of a homotopy as a continuous one-parameter family of maps from X to Y . If we imagine the parameter t as representing time, then the homotopy F represents a continuous “deforming” of the map f to the map f' , as t goes from 0 to 1.

Now we consider the special case in which f is a path in X . Recall that if $f : [0, 1] \rightarrow X$ is a continuous map such that $f(0) = x_0$ and $f(1) = x_1$, we say that f is a path in X from x_0 to x_1 . We also say that x_0 is the **initial point**, and x_1 the **final point**, of the path f . In this chapter, we shall for convenience use the interval $I = [0, 1]$ as the domain for all paths.

If f and f' are two paths in X , there is a stronger relation between them than mere homotopy. It is defined as follows:

Definition. Two paths f and f' , mapping the interval $I = [0, 1]$ into X , are said to be **path homotopic** if they have the same initial point x_0 and the same final point x_1 , and if there is a continuous map $F : I \times I \rightarrow X$ such that

$$\begin{aligned} F(s, 0) &= f(s) & \text{and} & & F(s, 1) &= f'(s), \\ F(0, t) &= x_0 & \text{and} & & F(1, t) &= x_1, \end{aligned}$$

for each $s \in I$ and each $t \in I$. We call F a **path homotopy** between f and f' . See Figure 51.1. If f is path homotopic to f' , we write $f \simeq_p f'$.

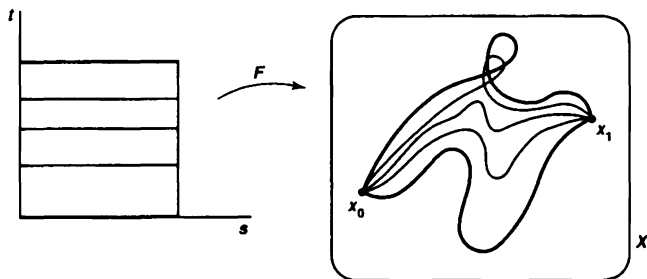


Figure 51.1

The first condition says simply that F is a homotopy between f and f' , and the second says that for each t , the path f_t defined by the equation $f_t(s) = F(s, t)$ is a path from x_0 to x_1 . Said differently, the first condition says that F represents a continuous way of deforming the path f to the path f' , and the second condition says that the end points of the path remain fixed during the deformation.

Lemma 51.1. *The relations $\simeq$ and $\simeq_p$ are equivalence relations.*

If f is a path, we shall denote its path-homotopy equivalence class by $[f]$.

Proof. Let us verify the properties of an equivalence relation.

Given f , it is trivial that $f \simeq f$; the map $F(x, t) = f(x)$ is the required homotopy. If f is a path, F is a path homotopy.

Given $f \simeq f'$, we show that $f' \simeq f$. Let F be a homotopy between f and f' . Then $G(x, t) = F(x, 1 - t)$ is a homotopy between f' and f . If F is a path homotopy, so is G .

Suppose that $f \simeq f'$ and $f' \simeq f''$. We show that $f \simeq f''$. Let F be a homotopy between f and f' , and let F' be a homotopy between f' and f'' . Define $G : X \times I \rightarrow Y$ by the equation

$$G(x, t) = \begin{cases} F(x, 2t) & \text{for } t \in [0, \frac{1}{2}], \\ F'(x, 2t - 1) & \text{for } t \in [\frac{1}{2}, 1]. \end{cases}$$

The map G is well defined, since if $t = \frac{1}{2}$, we have $F(x, 2t) = f'(x) = F'(x, 2t - 1)$. Because G is continuous on the two closed subsets $X \times [0, \frac{1}{2}]$ and $X \times [\frac{1}{2}, 1]$ of $X \times I$, it is continuous on all of $X \times I$, by the pasting lemma. Thus G is the required homotopy between f and f'' .

You can check that if F and F' are path homotopies, so is G . See Figure 51.2. ■

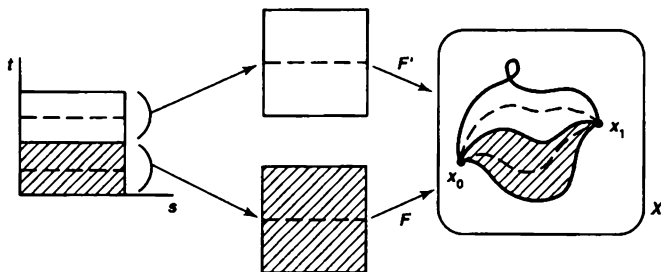


Figure 51.2

EXAMPLE 1. Let f and g be any two maps of a space X into $\mathbb{R}^2$. It is easy to see that f and g are homotopic; the map

$$F(x, t) = (1 - t)f(x) + tg(x)$$

is a homotopy between them. It is called a **straight-line homotopy** because it moves the point $f(x)$ to the point $g(x)$ along the straight-line segment joining them.

If f and g are paths from x_0 to x_1 , then F will be a path homotopy, as you can check. This situation is pictured in Figure 51.3.

More generally, let A be any *convex* subspace of $\mathbb{R}^n$. (This means that for any two points a, b of A , the straight line segment joining a and b is contained in A .) Then any two paths f, g in A from x_0 to x_1 are path homotopic in A , for the straight-line homotopy F between them has image set in A .

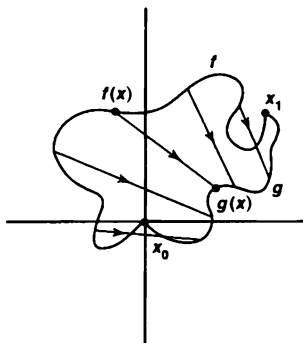


Figure 51.3

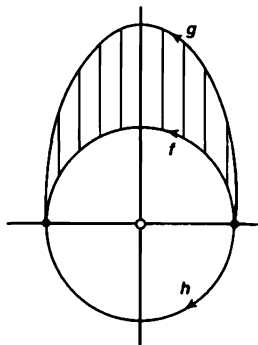


Figure 51.4

EXAMPLE 2 Let X denote the **punctured plane**, $\mathbb{R}^2 - \{0\}$, which we shall denote by $\mathbb{R}^2 - 0$ for short. The following paths in X ,

$$f(s) = (\cos \pi s, \sin \pi s),$$

$$g(s) = (\cos \pi s, 2 \sin \pi s)$$

are path homotopic, the straight-line homotopy between them is an acceptable path homotopy. But the straight-line homotopy between f and the path

$$h(s) = (\cos \pi s, -\sin \pi s)$$

is not acceptable, for its image does not lie in the space $X = \mathbb{R}^2 - 0$. See Figure 51.4.

Indeed, there exists *no* path homotopy in X between paths f and h . This result is hardly surprising, it is intuitively clear that one cannot “deform f past the hole at 0 ” without introducing a discontinuity. But it takes some work to prove. We shall return to this example later.

This example illustrates the fact that you must know what the range space is before you can tell whether two paths are path homotopic or not. The paths f and h would be path homotopic if they were paths in $\mathbb{R}^2$.

Now we introduce some algebra into this geometric situation. We define a certain operation on path-homotopy classes as follows:

Definition. If f is a path in X from x_0 to x_1 , and if g is a path in X from x_1 to x_2 , we define the **product** $f * g$ of f and g to be the path h given by the equations

$$h(s) = \begin{cases} f(2s) & \text{for } s \in [0, \frac{1}{2}], \\ g(2s - 1) & \text{for } s \in [\frac{1}{2}, 1]. \end{cases}$$

The function h is well-defined and continuous, by the pasting lemma; it is a path in X from x_0 to x_2 . We think of h as the path whose first half is the path f and whose second half is the path g .

The product operation on paths induces a well-defined operation on path-homotopy classes, defined by the equation

$$[f] * [g] = [f * g]$$

To verify this fact, let F be a path homotopy between f and f' and let G be a path homotopy between g and g' . Define

$$H(s, t) = \begin{cases} F(2s, t) & \text{for } s \in [0, \frac{1}{2}], \\ G(2s - 1, t) & \text{for } s \in [\frac{1}{2}, 1]. \end{cases}$$

Because $F(1, t) = x_1 = G(0, t)$ for all t , the map H is well-defined; it is continuous by the pasting lemma. You can check that H is the required path homotopy between $f * g$ and $f' * g'$. It is pictured in Figure 51.5.

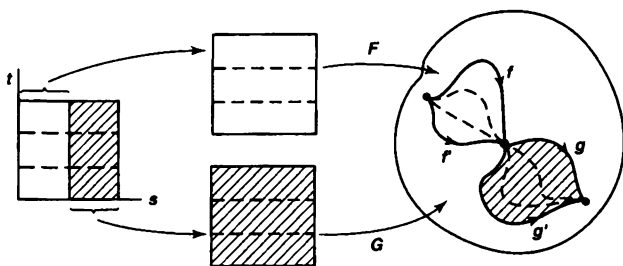


Figure 51.5

The operation $*$ on path-homotopy classes turns out to satisfy properties that look very much like the axioms for a group. They are called the *groupoid properties* of $*$. One difference from the properties of a group is that $[f] * [g]$ is not defined for every pair of classes, but only for those pairs $[f], [g]$ for which $f(1) = g(0)$.

Theorem 51.2. The operation $*$ has the following properties:

- (1) (Associativity) If $[f] * ([g] * [h])$ is defined, so is $([f] * [g]) * [h]$, and they are equal.

(2) (Right and left identities) Given $x \in X$, let e_x denote the constant path $e_x : I \rightarrow X$ carrying all of I to the point x . If f is a path in X from x_0 to x_1 , then

$$[f] * [e_{x_1}] = [f] \quad \text{and} \quad [e_{x_0}] * [f] = [f].$$

(3) (Inverse) Given the path f in X from x_0 to x_1 , let $\bar{f}$ be the path defined by $\bar{f}(s) = f(1 - s)$. It is called the *reverse* of f . Then

$$[f] * [\bar{f}] = [e_{x_0}] \quad \text{and} \quad [\bar{f}] * [f] = [e_{x_1}].$$

Proof. We shall make use of two elementary facts. The first is the fact that if $k : X \rightarrow Y$ is a continuous map, and if F is a path homotopy in X between the paths f and f' , then $k \circ F$ is a path homotopy in Y between the paths $k \circ f$ and $k \circ f'$. See Figure 51.6.



Figure 51.6

The second is the fact that if $k : X \rightarrow Y$ is a continuous map and if f and g are paths in X with $f(1) = g(0)$, then

$$k \circ (f * g) = (k \circ f) * (k \circ g).$$

This equation follows at once from the definition of the product operation $*$.

Step 1. We verify properties (2) and (3). To verify (2), we let e_0 denote the constant path in I at 0, and we let $i : I \rightarrow I$ denote the identity map, which is a path in I from 0 to 1. Then $e_0 * i$ is also a path in I from 0 to 1. (The graphs of these two paths are pictured in Figure 51.7.)

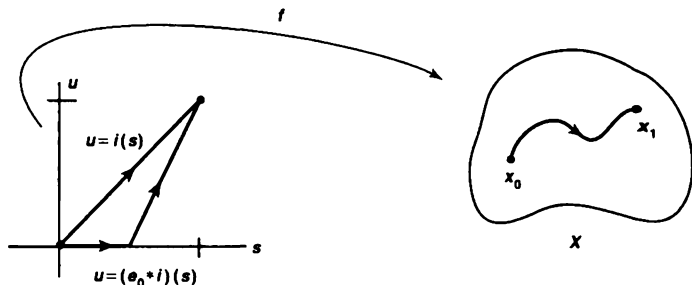


Figure 51.7

Because I is convex, there is a path homotopy G in I between i and $e_0 * i$. Then $f \circ G$ is a path homotopy in X between the paths $f \circ i = f$ and

$$f \circ (e_0 * i) = (f \circ e_0) * (f \circ i) = e_{x_0} * f.$$

An entirely similar argument, using the fact that if e_1 denotes the constant path at 1, then $i * e_1$ is path homotopic in I to the path i , shows that $[f] * [e_{x_1}] = [f]$.

To verify (3), note that the reverse of i is $\bar{i}(s) = 1 - s$. Then $i * \bar{i}$ is a path in I beginning and ending at 0, and so is the constant path e_0 . (Their graphs are pictured in Figure 51.8.) Because I is convex, there is a path homotopy H in I between e_0 and $i * \bar{i}$. Then $f \circ H$ is a path homotopy between $f \circ e_0 = e_{x_0}$ and

$$(f \circ i) * (f \circ \bar{i}) = f * \bar{f}.$$

An entirely similar argument, using the fact that $\bar{i} * i$ is path homotopic in I to e_1 , shows that $[\bar{f}] * [f] = [e_{x_1}]$.

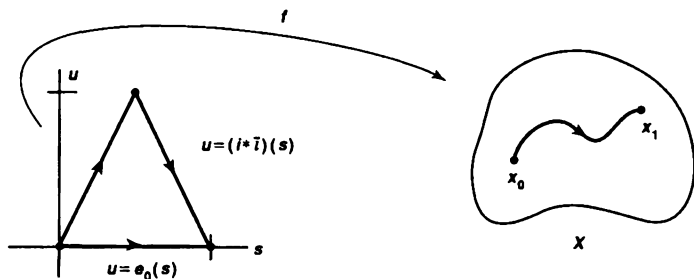


Figure 51.8

Step 2. The proof of (1), associativity, is a bit trickier. For this proof, and for later use as well, it will be convenient to describe the product $f * g$ in a different way.

If $[a, b]$ and $[c, d]$ are two intervals in $\mathbb{R}$, there is a unique map $p: [a, b] \rightarrow [c, d]$ of the form $p(x) = mx + k$ that carries a to c and b to d ; we call it the **positive linear map** of $[a, b]$ to $[c, d]$ because its graph is a straight line with positive slope. Note that the inverse of such a map is another such map, and so is the composite of two such maps.

With this terminology, the product $f * g$ can be described as follows: On $[0, \frac{1}{2}]$, it equals the positive linear map of $[0, \frac{1}{2}]$ to $[0, 1]$, followed by f ; and on $[\frac{1}{2}, 1]$, it equals the positive linear map of $[\frac{1}{2}, 1]$ to $[0, 1]$, followed by g .

Now we verify (1). Given paths f , g , and h in X , the products $f * (g * h)$ and $(f * g) * h$ are defined precisely when $f(1) = g(0)$ and $g(1) = h(0)$. Assuming these two conditions, we define also a "triple product" of the paths f , g , and h as follows: Choose points a and b of I so that $0 < a < b < 1$. Define a path $k_{a,b}$ in X as follows:

On $[0, a]$ it equals the positive linear map of $[0, a]$ to I followed by f ; on $[a, b]$ it equals the positive linear map of $[a, b]$ to I followed by g ; and on $[b, 1]$ it equals the positive linear map of $[b, 1]$ to I followed by h . The path $k_{a,b}$ depends of course on the choice of the points a and b . But its path-homotopy class does not! We show that if c and d are another pair of points of I with $0 < c < d < 1$, then $k_{c,d}$ is path homotopic to $k_{a,b}$.

Let $p : I \rightarrow I$ be the map whose graph is pictured in Figure 51.9. When restricted to $[0, a]$, $[a, b]$, and $[b, 1]$, respectively, it equals the positive linear maps of these intervals onto $[0, c]$, $[c, d]$, and $[d, 1]$, respectively. It follows at once that $k_{c,d} \circ p$ equals $k_{a,b}$. But p is a path in I from 0 to 1; and so is the identity map $i : I \rightarrow I$. Hence, there is a path homotopy P in I between p and i . Then $k_{c,d} \circ P$ is a path homotopy in X between $k_{a,b}$ and $k_{c,d}$.

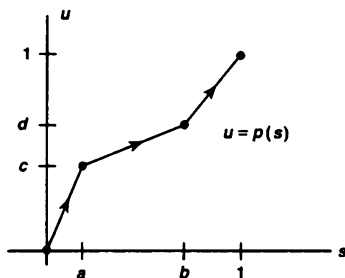


Figure 51.9

What has this to do with associativity? A great deal. For the product $f * (g * h)$ is exactly the triple product $k_{a,b}$ in the case where $a = 1/2$ and $b = 3/4$, as you can check, while the product $(f * g) * h$ equals $k_{c,d}$ in the case where $c = 1/4$ and $d = 1/2$. Therefore these two products are path homotopic. ■

The argument just used to prove associativity goes through for any finite product of paths. Roughly speaking, it says that as far as the path-homotopy class of the result is concerned, it doesn't matter how you chop up the interval when you form the product of paths! This result will be useful to us later, so we state it formally as a theorem here:

Theorem 51.3. Let f be a path in X , and let $a_0, \dots, a_n$ be numbers such that $0 = a_0 < a_1 < \dots < a_n = 1$. Let $f_i : I \rightarrow X$ be the path that equals the positive linear map of I onto $[a_{i-1}, a_i]$ followed by f . Then

$$[f] = [f_1] * \dots * [f_n].$$

Exercises

1. Show that if $h, h' : X \rightarrow Y$ are homotopic and $k, k' : Y \rightarrow Z$ are homotopic, then $k \circ h$ and $k' \circ h'$ are homotopic.
2. Given spaces X and Y , let $[X, Y]$ denote the set of homotopy classes of maps of X into Y .
 - (a) Let $I = [0, 1]$. Show that for any X , the set $[X, I]$ has a single element.
 - (b) Show that if Y is path connected, the set $[I, Y]$ has a single element.
3. A space X is said to be **contractible** if the identity map $i_X : X \rightarrow X$ is nullhomotopic.
 - (a) Show that I and $\mathbb{R}$ are contractible.
 - (b) Show that a contractible space is path connected.
 - (c) Show that if Y is contractible, then for any X , the set $[X, Y]$ has a single element.
 - (d) Show that if X is contractible and Y is path connected, then $[X, Y]$ has a single element.

§52 The Fundamental Group

The set of path-homotopy classes of paths in a space X does not form a group under the operation $*$ because the product of two path-homotopy classes is not always defined. But suppose we pick out a point x_0 of X to serve as a “base point” and restrict ourselves to those paths that begin and end at x_0 . The set of these path-homotopy classes does form a group under $*$. It will be called the *fundamental group* of X .

In this section, we shall study the fundamental group and derive some of its properties. In particular, we shall show that the group is a topological invariant of the space X , the fact that is of crucial importance in using it to study homeomorphism problems.

Let us first review some terminology from group theory. Suppose G and G' are groups, written multiplicatively. A **homomorphism** $f : G \rightarrow G'$ is a map such that $f(x \cdot y) = f(x) \cdot f(y)$ for all x, y ; it automatically satisfies the equations $f(e) = e'$ and $f(x^{-1}) = f(x)^{-1}$, where e and e' are the identities of G and G' , respectively, and the exponent -1 denotes the inverse. The **kernel** of f is the set $f^{-1}(e')$; it is a subgroup of G . The image of f , similarly, is a subgroup of G' . The homomorphism f is called a **monomorphism** if it is injective (or equivalently, if the kernel of f consists of e alone). It is called an **epimorphism** if it is surjective; and it is called an **isomorphism** if it is bijective.

Suppose G is a group and H is a subgroup of G . Let xH denote the set of all products xh , for $h \in H$; it is called a **left coset** of H in G . The collection of all such cosets forms a partition of G . Similarly, the collection of all right cosets Hx of H in G forms a partition of G . We call H a **normal subgroup** of G if $x \cdot h \cdot x^{-1} \in H$ for each $x \in G$ and each $h \in H$. In this case, we have $xH = Hx$ for each x , so that our two

partitions of G are the same. We denote this partition by G/H ; if one defines

$$(xH) \cdot (yH) = (x \cdot y)H,$$

one obtains a well-defined operation on G/H that makes it a group. This group is called the *quotient* of G by H . The map $f : G \rightarrow G/H$ carrying x to xH is an epimorphism with kernel H . Conversely, if $f : G \rightarrow G'$ is an epimorphism, then its kernel N is a normal subgroup of G , and f induces an isomorphism $G/N \rightarrow G'$ that carries xN to $f(x)$ for each $x \in G$.

If the subgroup H of G is not normal, it will still be convenient to use the symbol G/H ; we will use it to denote the collection of *right* cosets of H in G .

Now we define the fundamental group.

Definition. Let X be a space; let x_0 be a point of X . A path in X that begins and ends at x_0 is called a *loop* based at x_0 . The set of path homotopy classes of loops based at x_0 , with the operation $*$, is called the *fundamental group* of X relative to the *base point* x_0 . It is denoted by $\pi_1(X, x_0)$.

It follows from Theorem 51.2 that the operation $*$, when restricted to this set, satisfies the axioms for a group. Given two loops f and g based at x_0 , the product $f * g$ is always defined and is a loop based at x_0 . Associativity, the existence of an identity element $[e_{x_0}]$, and the existence of an inverse $[\bar{f}]$ for $[f]$ are immediate.

Sometimes this group is called the *first homotopy group* of X , which term implies that there is a second homotopy group. There are indeed groups $\pi_n(X, x_0)$ for all $n \in \mathbb{Z}_+$, but we shall not study them in this book. They are part of the general subject called *homotopy theory*.

EXAMPLE 1 Let $\mathbb{R}^n$ denote euclidean n -space. Then $\pi_1(\mathbb{R}^n, x_0)$ is the trivial group (the group consisting of the identity alone). For if f is a loop in $\mathbb{R}^n$ based at x_0 , the straight-line homotopy is a path homotopy between f and the constant path at x_0 . More generally, if X is any convex subset of $\mathbb{R}^n$, then $\pi_1(X, x_0)$ is the trivial group. In particular, the *unit ball* B^n in $\mathbb{R}^n$,

$$B^n = \{x \mid x_1^2 + \cdots + x_n^2 \leq 1\},$$

has trivial fundamental group.

An immediate question one asks is the extent to which the fundamental group depends on the base point. We consider that question now.

Definition. Let α be a path in X from x_0 to x_1 . We define a map

$$\hat{\alpha} : \pi_1(X, x_0) \longrightarrow \pi_1(X, x_1)$$

by the equation

$$\hat{\alpha}([f]) = [\bar{\alpha}] * [f] * [\alpha].$$

The map $\hat{\alpha}$, which we call " α -hat," is well-defined because the operation $*$ is well-defined. If f is a loop based at x_0 , then $\bar{\alpha} * (f * \alpha)$ is a loop based at x_1 . Hence $\hat{\alpha}$ maps $\pi_1(X, x_0)$ into $\pi_1(X, x_1)$, as desired; note that it depends only on the path-homotopy class of α . It is pictured in Figure 52.1.

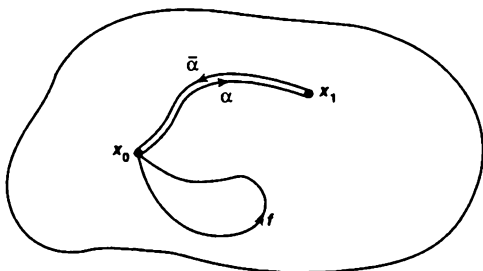


Figure 52.1

Theorem 52.1. The map $\hat{\alpha}$ is a group isomorphism.

Proof. To show that $\hat{\alpha}$ is a homomorphism, we compute

$$\begin{aligned}\hat{\alpha}([f]) * \hat{\alpha}([g]) &= ([\bar{\alpha}] * [f] * [\alpha]) * ([\bar{\alpha}] * [g] * [\alpha]) \\ &= [\bar{\alpha}] * [f] * [g] * [\alpha] \\ &= \hat{\alpha}([f] * [g]).\end{aligned}$$

To show that $\hat{\alpha}$ is an isomorphism, we show that if β denotes the path $\bar{\alpha}$, which is the reverse of α , then $\hat{\beta}$ is an inverse for $\hat{\alpha}$. We compute, for each element $[h] \in \pi_1(X, x_1)$,

$$\begin{aligned}\hat{\beta}([h]) &= [\bar{\beta}] * [h] * [\beta] = [\alpha] * [h] * [\bar{\alpha}], \\ \hat{\alpha}(\hat{\beta}([h])) &= [\bar{\alpha}] * ([\alpha] * [h] * [\bar{\alpha}]) * [\alpha] = [h].\end{aligned}$$

A similar computation shows that $\hat{\beta}(\hat{\alpha}([f])) = [f]$ for each $[f] \in \pi_1(X, x_0)$. ■

Corollary 52.2. If X is path connected and x_0 and x_1 are two points of X , then $\pi_1(X, x_0)$ is isomorphic to $\pi_1(X, x_1)$.

Suppose that X is a topological space. Let C be the path component of X containing x_0 . It is easy to see that $\pi_1(C, x_0) = \pi_1(X, x_0)$, since all loops and homotopies in X that are based at x_0 must lie in the subspace C . Thus $\pi_1(X, x_0)$ depends on only the path component of X containing x_0 ; it gives us no information whatever about the rest of X . For this reason, it is usual to deal with only path-connected spaces when studying the fundamental group.

If X is path connected, all the groups $\pi_1(X, x)$ are isomorphic, so it is tempting to try to “identify” all these groups with one another and to speak simply of the fundamental group of the space X , without reference to base point. The difficulty with this approach is that there is no *natural* way of identifying $\pi_1(X, x_0)$ with $\pi_1(X, x_1)$; different paths α and β from x_0 to x_1 may give rise to different isomorphisms between these groups. For this reason, omitting the base point can lead to error.

It turns out that the isomorphism of $\pi_1(X, x_0)$ with $\pi_1(X, x_1)$ is independent of path if and only if the fundamental group is abelian. (See Exercise 3.) This is a stringent requirement on the space X .

Definition. A space X is said to be *simply connected* if it is a path-connected space and if $\pi_1(X, x_0)$ is the trivial (one-element) group for some $x_0 \in X$, and hence for every $x_0 \in X$. We often express the fact that $\pi_1(X, x_0)$ is the trivial group by writing $\pi_1(X, x_0) = 0$.

Lemma 52.3. *In a simply connected space X , any two paths having the same initial and final points are path homotopic.*

Proof. Let α and β be two paths from x_0 to x_1 . Then $\alpha * \bar{\beta}$ is defined and is a loop on X based at x_0 . Since X is simply connected, this loop is path homotopic to the constant loop at x_0 . Then

$$[\alpha * \bar{\beta}] * [\beta] = [e_{x_0}] * [\beta],$$

from which it follows that $[\alpha] = [\beta]$. ■

It is intuitively clear that the fundamental group is a topological invariant of the space X . A convenient way to prove this fact formally is to introduce the notion of the "homomorphism induced by a continuous map."

Suppose that $h : X \rightarrow Y$ is a continuous map that carries the point x_0 of X to the point y_0 of Y . We often denote this fact by writing

$$h : (X, x_0) \longrightarrow (Y, y_0).$$

If f is a loop in X based at x_0 , then the composite $h \circ f : I \rightarrow Y$ is a loop in Y based at y_0 . The correspondence $f \rightarrow h \circ f$ thus gives rise to a map carrying $\pi_1(X, x_0)$ into $\pi_1(Y, y_0)$. We define it formally as follows:

Definition. Let $h : (X, x_0) \rightarrow (Y, y_0)$ be a continuous map. Define

$$h_* : \pi_1(X, x_0) \longrightarrow \pi_1(Y, y_0)$$

by the equation

$$h_*([f]) = [h \circ f].$$

The map h_* is called the *homomorphism induced by h* , relative to the base point x_0 .

The map h_* is well-defined, for if F is a path homotopy between the paths f and f' , then $h \circ F$ is a path homotopy between the paths $h \circ f$ and $h \circ f'$. The fact that h_* is a homomorphism follows from the equation

$$(h \circ f) * (h \circ g) = h \circ (f * g).$$

The homomorphism h_* depends not only on the map $h : X \rightarrow Y$ but also on the choice of the base point x_0 . (Once x_0 is chosen, y_0 is determined by h .) So some notational difficulty will arise if we want to consider several different base points for X . If x_0 and x_1 are two different points of X , we cannot use the same symbol h_* to stand for two different homomorphisms, one having domain $\pi_1(X, x_0)$ and the other having domain $\pi_1(X, x_1)$. Even if X is path connected, so these groups are isomorphic, they are still not the same group. In such a case, we shall use the notation

$$(h_{x_0})_* : \pi_1(X, x_0) \longrightarrow \pi_1(Y, y_0)$$

for the first homomorphism and $(h_{x_1})_*$ for the second. If there is only one base point under consideration, we shall omit mention of the base point and denote the induced homomorphism merely by h_* .

The induced homomorphism has two properties that are crucial in the applications. They are called its "functorial properties" and are given in the following theorem:

Theorem 52.4. *If $h : (X, x_0) \rightarrow (Y, y_0)$ and $k : (Y, y_0) \rightarrow (Z, z_0)$ are continuous, then $(k \circ h)_* = k_* \circ h_*$. If $i : (X, x_0) \rightarrow (X, x_0)$ is the identity map, then i_* is the identity homomorphism.*

Proof. The proof is a triviality. By definition,

$$\begin{aligned}(k \circ h)_*([f]) &= [(k \circ h) \circ f], \\ (k_* \circ h_*)([f]) &= k_*(h_*([f])) = k_*([h \circ f]) = [k \circ (h \circ f)].\end{aligned}$$

Similarly, $i_*([f]) = [i \circ f] = [f]$ ■

Corollary 52.5. *If $h : (X, x_0) \rightarrow (Y, y_0)$ is a homeomorphism of X with Y , then h_* is an isomorphism of $\pi_1(X, x_0)$ with $\pi_1(Y, y_0)$.*

Proof. Let $k : (Y, y_0) \rightarrow (X, x_0)$ be the inverse of h . Then $k_* \circ h_* = (k \circ h)_* = i_*$, where i is the identity map of (X, x_0) ; and $h_* \circ k_* = (h \circ k)_* = j_*$, where j is the identity map of (Y, y_0) . Since i_* and j_* are the identity homomorphisms of the groups $\pi_1(X, x_0)$ and $\pi_1(Y, y_0)$, respectively, k_* is the inverse of h_* . ■

Exercises

- A subset A of $\mathbb{R}^n$ is said to be **star convex** if for some point a_0 of A , all the line segments joining a_0 to other points of A lie in A .
 - Find a star convex set that is not convex.
 - Show that if A is star convex, A is simply connected.
- Let α be a path in X from x_0 to x_1 ; let β be a path in X from x_1 to x_2 . Show that if $\gamma = \alpha * \beta$, then $\hat{\gamma} = \hat{\beta} \circ \hat{\alpha}$.

- Let x_0 and x_1 be points of the path-connected space X . Show that $\pi_1(X, x_0)$ is abelian if and only if for every pair α and β of paths from x_0 to x_1 , we have $\hat{\alpha} = \hat{\beta}$.
- Let $A \subset X$; suppose $r : X \rightarrow A$ is a continuous map such that $r(a) = a$ for each $a \in A$. (The map r is called a **retraction** of X onto A .) If $a_0 \in A$, show that

$$r_* : \pi_1(X, a_0) \longrightarrow \pi_1(A, a_0)$$

is surjective.

- Let A be a subspace of $\mathbb{R}^n$; let $h : (A, a_0) \rightarrow (Y, y_0)$. Show that if h is extendable to a continuous map of $\mathbb{R}^n$ into Y , then h_* is the trivial homomorphism (the homomorphism that maps everything to the identity element).
- Show that if X is path connected, the homomorphism induced by a continuous map is independent of base point, up to isomorphisms of the groups involved. More precisely, let $h : X \rightarrow Y$ be continuous, with $h(x_0) = y_0$ and $h(x_1) = y_1$. Let α be a path in X from x_0 to x_1 , and let $\beta = h \circ \alpha$. Show that

$$\hat{\beta} \circ (h_{x_0})_* = (h_{x_1})_* \circ \hat{\alpha}.$$

This equation expresses the fact that the following diagram of maps “commutes.”

$$\begin{array}{ccc} \pi_1(X, x_0) & \xrightarrow{(h_{x_0})_*} & \pi_1(Y, y_0) \\ \downarrow \hat{\alpha} & & \downarrow \hat{\beta} \\ \pi_1(X, x_1) & \xrightarrow{(h_{x_1})_*} & \pi_1(Y, y_1) \end{array}$$

- Let G be a topological group with operation $\cdot$ and identity element x_0 . Let $\Omega(G, x_0)$ denote the set of all loops in G based at x_0 . If $f, g \in \Omega(G, x_0)$, let us define a loop $f \otimes g$ by the rule

$$(f \otimes g)(s) = f(s) \cdot g(s).$$

- Show that this operation makes the set $\Omega(G, x_0)$ into a group.
 - Show that this operation induces a group operation $\otimes$ on $\pi_1(G, x_0)$.
 - Show that the two group operations $*$ and $\otimes$ on $\pi_1(G, x_0)$ are the same.
- [Hint: Compute $(f * e_{x_0}) \otimes (e_{x_0} * g)$]
- Show that $\pi_1(G, x_0)$ is abelian.

§53 Covering Spaces

We have shown that any convex subspace of $\mathbb{R}^n$ has a trivial fundamental group; we turn now to the task of computing some fundamental groups that are not trivial. One of the most useful tools for this purpose is the notion of *covering space*, which we introduce in this section. Covering spaces are also important in the study of Riemann surfaces and complex manifolds. (See [A-S].) We shall study them in more detail in Chapter 13.

Definition. Let $p: E \rightarrow B$ be a continuous surjective map. The open set U of B is said to be **evenly covered** by p if the inverse image $p^{-1}(U)$ can be written as the union of disjoint open sets V_α in E such that for each α , the restriction of p to V_α is a homeomorphism of V_α onto U . The collection $\{V_\alpha\}$ will be called a **partition of $p^{-1}(U)$ into slices**

If U is an open set that is evenly covered by p , we often picture the set $p^{-1}(U)$ as a “stack of pancakes,” each having the same size and shape as U , floating in the air above U ; the map p squashes them all down onto U . See Figure 53.1. Note that if U is evenly covered by p and W is an open set contained in U , then W is also evenly covered by p .

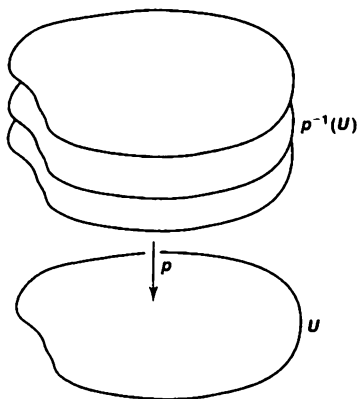


Figure 53.1

Definition. Let $p: E \rightarrow B$ be continuous and surjective. If every point b of B has a neighborhood U that is evenly covered by p , then p is called a **covering map**, and E is said to be a **covering space of B**

Note that if $p: E \rightarrow B$ is a covering map, then for each $b \in B$ the subspace $p^{-1}(b)$ of E has the discrete topology. For each slice V_α is open in E and intersects the set $p^{-1}(b)$ in a single point; therefore, this point is open in $p^{-1}(b)$.

Note also that if $p: E \rightarrow B$ is a covering map, then p is an open map. For suppose A is an open set of E . Given $x \in p(A)$, choose a neighborhood U of x that is evenly covered by p . Let $\{V_\alpha\}$ be a partition of $p^{-1}(U)$ into slices. There is a point y of A such that $p(y) = x$; let V_β be the slice containing y . The set $V_\beta \cap A$ is open in E and hence open in V_β , because p maps V_β homeomorphically onto U , the set $p(V_\beta \cap A)$ is open in U and hence open in B ; it is thus a neighborhood of x contained in $p(A)$, as desired.

EXAMPLE 1. Let X be any space; let $i : X \rightarrow X$ be the identity map. Then i is a covering map (of the most trivial sort). More generally, let E be the space $X \times \{1, \dots, n\}$ consisting of n disjoint copies of X . The map $p : E \rightarrow X$ given by $p(x, i) = x$ for all i is again a (rather trivial) covering map. In this case, we can picture the entire space E as a stack of pancakes over X .

In practice, one often restricts oneself to covering spaces that are path connected, to eliminate trivial coverings of the pancake-stack variety. An example of such a non-trivial covering space is the following.

Theorem 53.1. The map $p : \mathbb{R} \rightarrow S^1$ given by the equation

$$p(x) = (\cos 2\pi x, \sin 2\pi x)$$

is a covering map.

One can picture p as a function that wraps the real line $\mathbb{R}$ around the circle S^1 , and in the process maps each interval $[n, n+1]$ onto S^1 .

Proof. The fact that p is a covering map comes from elementary properties of the sine and cosine functions. Consider, for example, the subset U of S^1 consisting of those points having positive first coordinate. The set $p^{-1}(U)$ consists of those points x for which $\cos 2\pi x$ is positive, that is, it is the union of the intervals

$$V_n = (n - \frac{1}{4}, n + \frac{1}{4}),$$

for all $n \in \mathbb{Z}$. See Figure 53.2. Now, restricted to any closed interval $\bar{V}_n$, the map p is injective because $\sin 2\pi x$ is strictly monotonic on such an interval. Furthermore, p carries $\bar{V}_n$ surjectively onto $\bar{U}$, and V_n to U , by the intermediate value theorem. Since $\bar{V}_n$ is compact, $p|_{\bar{V}_n}$ is a homeomorphism of $\bar{V}_n$ with $\bar{U}$. In particular, $p|_{V_n}$ is a homeomorphism of V_n with U .

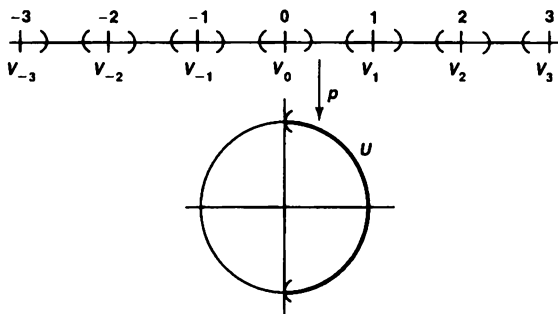


Figure 53.2

Similar arguments can be applied to the intersections of S^1 with the upper and lower open half-planes, and with the open left-hand half-plane. These open sets

cover S^1 , and each of them is evenly covered by p . Hence $p : \mathbb{R} \rightarrow S^1$ is a covering map. ■

If $p : E \rightarrow B$ is a covering map, then p is a **local homeomorphism** of E with B . That is, each point e of E has a neighborhood that is mapped homeomorphically by p onto an open subset of B . The condition that p be a local homeomorphism does not suffice, however, to ensure that p is a covering map, as the following example shows.

EXAMPLE 2. The map $p : \mathbb{R}_+ \rightarrow S^1$ given by the equation

$$p(x) = (\cos 2\pi x, \sin 2\pi x)$$

is surjective, and it is a local homeomorphism. See Figure 53.3. But it is not a covering map, for the point $b_0 = (1, 0)$ has no neighborhood U that is evenly covered by p . The typical neighborhood U of b_0 has an inverse image consisting of small neighborhoods V_n of each integer n for $n > 0$, along with a small interval V_0 of the form $(0, \epsilon)$. Each of the intervals V_n for $n > 0$ is mapped homeomorphically onto U by the map p , but the interval V_0 is only imbedded in U by p .

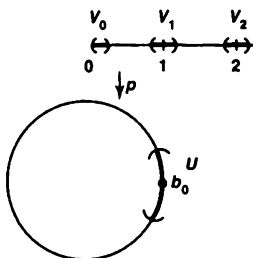


Figure 53.3

EXAMPLE 3. The preceding example might lead you to think that the real line $\mathbb{R}$ is the only connected covering space of the circle S^1 . This is not so. Consider, for example, the map $p : S^1 \rightarrow S^1$ given in equations by

$$p(z) = z^2.$$

[Here we consider S^1 as the subset of the complex plane $\mathbb{C}$ consisting of those complex numbers z with $|z| = 1$] We leave it to you to check that p is a covering map.

Example 2 shows that the map obtained by restricting a covering map may not be a covering map. Here is one situation where it *will* be a covering map:

Theorem 53.2. Let $p : E \rightarrow B$ be a covering map. If B_0 is a subspace of B , and if $E_0 = p^{-1}(B_0)$, then the map $p_0 : E_0 \rightarrow B_0$ obtained by restricting p is a covering map.

Proof. Given $b_0 \in B_0$, let U be an open set in B containing b_0 that is evenly covered by p ; let $\{V_\alpha\}$ be a partition of $p^{-1}(U)$ into slices. Then $U \cap B_0$ is a neighborhood of b_0 in B_0 , and the sets $V_\alpha \cap E_0$ are disjoint open sets in E_0 whose union is $p^{-1}(U \cap B_0)$, and each is mapped homeomorphically onto $U \cap B_0$ by p . ■

Theorem 53.3. If $p : E \rightarrow B$ and $p' : E' \rightarrow B'$ are covering maps, then

$$p \times p' : E \times E' \rightarrow B \times B'$$

is a covering map.

Proof. Given $b \in B$ and $b' \in B'$, let U and U' be neighborhoods of b and b' , respectively, that are evenly covered by p and p' , respectively. Let $\{V_\alpha\}$ and $\{V'_\beta\}$ be partitions of $p^{-1}(U)$ and $(p')^{-1}(U')$, respectively, into slices. Then the inverse image under $p \times p'$ of the open set $U \times U'$ is the union of all the sets $V_\alpha \times V'_\beta$. These are disjoint open sets of $E \times E'$, and each is mapped homeomorphically onto $U \times U'$ by $p \times p'$. ■

EXAMPLE 4. Consider the space $T = S^1 \times S^1$, it is called the *torus*. The product map

$$p \times p : \mathbb{R} \times \mathbb{R} \rightarrow S^1 \times S^1$$

is a covering of the torus by the plane $\mathbb{R}^2$, where p denotes the covering map of Theorem 53.1. Each of the unit squares $[n, n+1] \times [m, m+1]$ gets wrapped by $p \times p$ entirely around the torus. See Figure 53.4

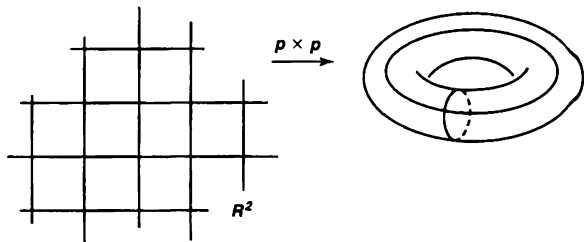


Figure 53.4

In this figure, we have pictured the torus not as the product $S^1 \times S^1$, which is a subspace of $\mathbb{R}^4$ and thus difficult to visualize, but as the familiar doughnut-shaped surface D in $\mathbb{R}^3$ obtained by rotating the circle C_1 in the xz -plane of radius $\frac{1}{2}$ centered at $(1, 0, 0)$ about the z -axis. It is not hard to see that $S^1 \times S^1$ is homeomorphic with the surface D . Let C_2 be the circle of radius 1 in the xy -plane centered at the origin. Then let us map $C_1 \times C_2$ into D by defining $f(a \times b)$ to be that point into which a is carried when one rotates the circle C_1 about the z -axis until its center hits the point b . See Figure 53.5. The map f will be a homeomorphism of $C_1 \times C_2$ with D , as you can check mentally. If you wish, you can write equations for f and check continuity, injectivity, and surjectivity directly. (Continuity of f^{-1} will follow from compactness of $C_1 \times C_2$.)

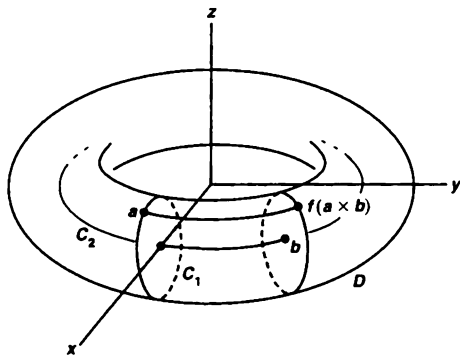


Figure 53.5

EXAMPLE 5. Consider the covering map $p \times p$ of the preceding example. Let b_0 denote the point $p(0)$ of S^1 ; and let B_0 denote the subspace

$$B_0 = (S^1 \times b_0) \cup (b_0 \times S^1)$$

of $S^1 \times S^1$. Then B_0 is the union of two circles that have a point in common, we sometimes call it the *figure-eight space*. The space $E_0 = p^{-1}(B_0)$ is the "infinite grid"

$$E_0 = (\mathbb{R} \times \mathbb{Z}) \cup (\mathbb{Z} \times \mathbb{R})$$

pictured in Figure 53.4. The map $p_0 : E_0 \rightarrow B_0$ obtained by restricting $p \times p$ is thus a covering map.

The infinite grid is but one covering space of the figure eight; we shall see others later on.

EXAMPLE 6 Consider the covering map

$$p \times i : \mathbb{R} \times \mathbb{R}_+ \longrightarrow S^1 \times \mathbb{R}_+,$$

where i is the identity map of $\mathbb{R}_+$ and p is the map of Theorem 53.1. If we take the standard homeomorphism of $S^1 \times \mathbb{R}_+$ with $\mathbb{R}^2 - 0$, sending $x \times t$ to tx , the composite gives us a covering

$$\mathbb{R} \times \mathbb{R}_+ \longrightarrow \mathbb{R}^2 - 0$$

of the punctured plane by the open upper half-plane. It is pictured in Figure 53.6. This covering map appears in the study of complex variables as the *Riemann surface* corresponding to the complex logarithm function.

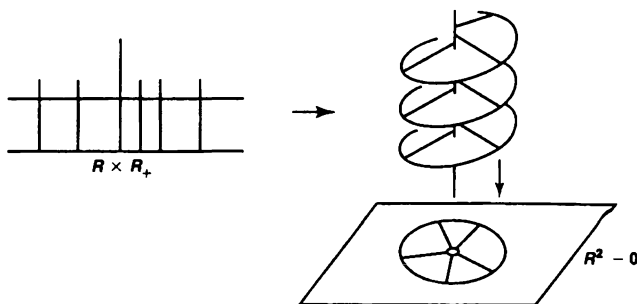


Figure 53.6

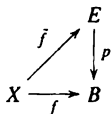
Exercises

1. Let Y have the discrete topology. Show that if $p : X \times Y \rightarrow X$ is projection on the first coordinate, then p is a covering map.
2. Let $p : E \rightarrow B$ be continuous and surjective. Suppose that U is an open set of B that is evenly covered by p . Show that if U is connected, then the partition of $p^{-1}(U)$ into slices is unique.
3. Let $p : E \rightarrow B$ be a covering map; let B be connected. Show that if $p^{-1}(b_0)$ has k elements for some $b_0 \in B$, then $p^{-1}(b)$ has k elements for every $b \in B$. In such a case, E is called a **k -fold covering** of B .
4. Let $q : X \rightarrow Y$ and $r : Y \rightarrow Z$ be covering maps; let $p = r \circ q$. Show that if $r^{-1}(z)$ is finite for each $z \in Z$, then p is a covering map.
5. Show that the map of Example 3 is a covering map. Generalize to the map $p(z) = z^n$.
6. Let $p : E \rightarrow B$ be a covering map.
 - (a) If B is Hausdorff, regular, completely regular, or locally compact Hausdorff, then so is E . [Hint: If $\{V_\alpha\}$ is a partition of $p^{-1}(U)$ into slices, and C is a closed set of B such that $C \subset U$, then $p^{-1}(C) \cap V_\alpha$ is a closed set of E .]
 - (b) If B is compact and $p^{-1}(b)$ is finite for each $b \in B$, then E is compact.

§54 The Fundamental Group of the Circle

The study of covering spaces of a space X is intimately related to the study of the fundamental group of X . In this section, we establish the crucial links between the two concepts, and compute the fundamental group of the circle.

Definition. Let $p : E \rightarrow B$ be a map. If f is a continuous mapping of some space X into B , a **lifting** of f is a map $\tilde{f} : X \rightarrow E$ such that $p \circ \tilde{f} = f$.



The existence of liftings when p is a covering map is an important tool in studying covering spaces and the fundamental group. First, we show that for a covering space, paths can be lifted; then we show that path homotopies can be lifted as well. First, an example:

EXAMPLE 1. Consider the covering $p : \mathbb{R} \rightarrow S^1$ of Theorem 53.1. The path $f : [0, 1] \rightarrow S^1$ beginning at $b_0 = (1, 0)$ given by $f(s) = (\cos \pi s, \sin \pi s)$ lifts to the path $\tilde{f}(s) = s/2$ beginning at 0 and ending at $\frac{1}{2}$. The path $g(s) = (\cos \pi s, -\sin \pi s)$ lifts to the path $\tilde{g}(s) = -s/2$ beginning at 0 and ending at $-\frac{1}{2}$. The path $h(s) = (\cos 4\pi s, \sin 4\pi s)$ lifts to the path $\tilde{h}(s) = 2s$ beginning at 0 and ending at 2. Intuitively, h wraps the interval $[0, 1]$ around the circle twice; this is reflected in the fact that the lifted path $\tilde{h}$ begins at zero and ends at the number 2. These paths are pictured in Figure 54.1.

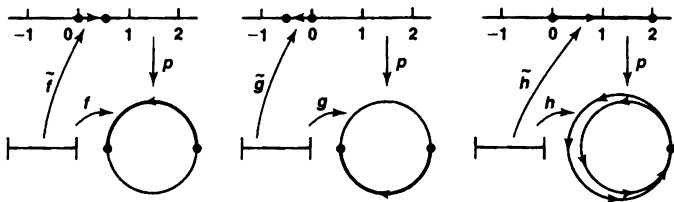


Figure 54.1

Lemma 54.1. Let $p : E \rightarrow B$ be a covering map, let $p(e_0) = b_0$. Any path $f : [0, 1] \rightarrow B$ beginning at b_0 has a unique lifting to a path $\tilde{f}$ in E beginning at e_0 .

Proof. Cover B by open sets U each of which is evenly covered by p . Find a subdivision of $[0, 1]$, say $s_0, \dots, s_n$, such that for each i the set $f([s_i, s_{i+1}])$ lies in such an open set U . (Here we use the Lebesgue number lemma.) We define the lifting $\tilde{f}$ step by step.

First, define $\tilde{f}(0) = e_0$. Then, supposing $\tilde{f}(s)$ is defined for $0 \leq s \leq s_i$, we define $\tilde{f}$ on $[s_i, s_{i+1}]$ as follows: The set $f([s_i, s_{i+1}])$ lies in some open set U that is evenly covered by p . Let $\{V_\alpha\}$ be a partition of $p^{-1}(U)$ into slices; each set V_α is mapped homeomorphically onto U by p . Now $\tilde{f}(s_i)$ lies in one of these sets, say in V_0 . Define $\tilde{f}(s)$ for $s \in [s_i, s_{i+1}]$ by the equation

$$\tilde{f}(s) = (p|_{V_0})^{-1}(f(s)).$$

Because $p|V_0 : V_0 \rightarrow U$ is a homeomorphism, $\tilde{f}$ will be continuous on $[s_i, s_{i+1}]$.

Continuing in this way, we define $\tilde{f}$ on all of $[0, 1]$. Continuity of $\tilde{f}$ follows from the pasting lemma; the fact that $p \circ \tilde{f} = f$ is immediate from the definition of $\tilde{f}$.

The uniqueness of $\tilde{f}$ is also proved step by step. Suppose that $\tilde{\tilde{f}}$ is another lifting of f beginning at e_0 . Then $\tilde{\tilde{f}}(0) = e_0 = \tilde{f}(0)$. Suppose that $\tilde{\tilde{f}}(s) = \tilde{f}(s)$ for all s such that $0 \leq s \leq s_i$. Let V_0 be as in the preceding paragraph; then for $s \in [s_i, s_{i+1}]$, $\tilde{\tilde{f}}(s)$ is defined as $(p|V_0)^{-1}(f(s))$. What can $\tilde{\tilde{f}}(s)$ equal? Since $\tilde{\tilde{f}}$ is a lifting of f , it must carry the interval $[s_i, s_{i+1}]$ into the set $p^{-1}(U) = \bigcup V_\alpha$. The slices V_α are open and disjoint; because the set $\tilde{\tilde{f}}([s_i, s_{i+1}])$ is connected, it must lie entirely in one of the sets V_α . Because $\tilde{\tilde{f}}(s_i) = \tilde{f}(s_i)$, which is in V_0 , $\tilde{\tilde{f}}$ must carry all of $[s_i, s_{i+1}]$ into the set V_0 . Thus, for s in $[s_i, s_{i+1}]$, $\tilde{\tilde{f}}(s)$ must equal some point y of V_0 lying in $p^{-1}(f(s))$. But there is only *one* such point y , namely, $(p|V_0)^{-1}(f(s))$. Hence $\tilde{\tilde{f}}(s) = \tilde{f}(s)$ for $s \in [s_i, s_{i+1}]$. ■

Lemma 54.2. Let $p : E \rightarrow B$ be a covering map; let $p(e_0) = b_0$. Let the map $F : I \times I \rightarrow B$ be continuous, with $F(0, 0) = b_0$. There is a unique lifting of F to a continuous map

$$\tilde{F} : I \times I \rightarrow E$$

such that $\tilde{F}(0, 0) = e_0$. If F is a path homotopy, then $\tilde{F}$ is a path homotopy.

Proof. Given F , we first define $\tilde{F}(0, 0) = e_0$. Next, we use the preceding lemma to extend $\tilde{F}$ to the left-hand edge $0 \times I$ and the bottom edge $I \times 0$ of $I \times I$. Then we extend $\tilde{F}$ to all of $I \times I$ as follows:

Choose subdivisions

$$\begin{aligned} s_0 < s_1 < \cdots < s_m, \\ t_0 < t_1 < \cdots < t_n \end{aligned}$$

of I fine enough that each rectangle

$$I_i \times J_j = [s_{i-1}, s_i] \times [t_{j-1}, t_j]$$

is mapped by F into an open set of B that is evenly covered by p . (Use the Lebesgue number lemma.) We define the lifting $\tilde{F}$ step by step, beginning with the rectangle $I_1 \times J_1$, continuing with the other rectangles $I_i \times J_1$ in the "bottom row," then with the rectangles $I_i \times J_2$ in the next row, and so on.

In general, given i_0 and j_0 , assume that $\tilde{F}$ is defined on the set A which is the union of $0 \times I$ and $I \times 0$ and all the rectangles "previous" to $I_{i_0} \times J_{j_0}$ (those rectangles $I_i \times J_j$ for which $j < j_0$ and those for which $j = j_0$ and $i < i_0$). Assume also that $\tilde{F}$ is a continuous lifting of $F|A$. We define $\tilde{F}$ on $I_{i_0} \times J_{j_0}$. Choose an open set U of B that is evenly covered by p and contains the set $F(I_{i_0} \times J_{j_0})$. Let $\{V_\alpha\}$ be a partition of $p^{-1}(U)$ into slices; each set V_α is mapped homeomorphically onto U by p . Now $\tilde{F}$ is already defined on the set $C = A \cap (I_{i_0} \times J_{j_0})$. This set is the union of the left

and bottom edges of the rectangle $I_{i_0} \times J_{j_0}$, so it is *connected*. Therefore, $\tilde{F}(C)$ is connected and must lie entirely within one of the sets V_α . Suppose it lies in V_0 . Then, the situation is as pictured in Figure 54.2.

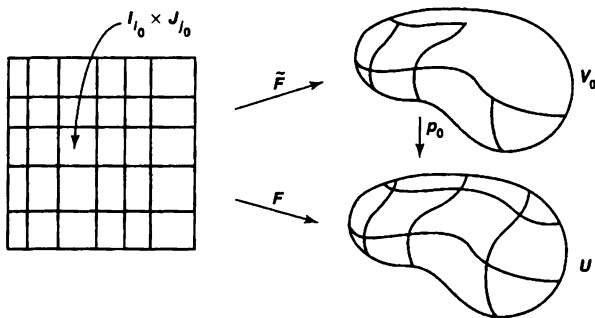


Figure 54.2

Let $p_0 : V_0 \rightarrow U$ denote the restriction of p to V_0 . Since $\tilde{F}$ is a lifting of $F|_A$, we know that for $x \in C$,

$$p_0(\tilde{F}(x)) = p(\tilde{F}(x)) = F(x),$$

so that $\tilde{F}(x) = p_0^{-1}(F(x))$. Hence we may extend $\tilde{F}$ by defining

$$\tilde{F}(x) = p_0^{-1}(F(x))$$

for $x \in I_{i_0} \times J_{j_0}$. The extended map will be continuous by the pasting lemma.

Continuing in this way, we define $\tilde{F}$ on all of I^2 .

To check uniqueness, note that at each step of the construction of $\tilde{F}$, as we extend $\tilde{F}$ first to the bottom and left edges of I^2 , and then to the rectangles $I_i \times J_j$, one by one, there is only one way to extend $\tilde{F}$ continuously. Thus, once the value of $\tilde{F}$ at $(0, 0)$ is specified, $\tilde{F}$ is completely determined.

Now suppose that F is a path homotopy. We wish to show that $\tilde{F}$ is a path homotopy. The map F carries the entire left edge $0 \times I$ of I^2 into a single point b_0 of B . Because $\tilde{F}$ is a lifting of F , it carries this edge into the set $p^{-1}(b_0)$. But this set has the discrete topology as a subspace of E . Since $0 \times I$ is connected and $\tilde{F}$ is continuous, $\tilde{F}(0 \times I)$ is connected and thus must equal a one-point set. Similarly, $\tilde{F}(1 \times I)$ must be a one-point set. Thus $\tilde{F}$ is a path homotopy. ■

Theorem 54.3. Let $p : E \rightarrow B$ be a covering map; let $p(e_0) = b_0$. Let f and g be two paths in B from b_0 to b_1 , let $\tilde{f}$ and $\tilde{g}$ be their respective liftings to paths in E beginning at e_0 . If f and g are path homotopic, then $\tilde{f}$ and $\tilde{g}$ end at the same point of E and are path homotopic.

Proof. Let $F : I \times I \rightarrow B$ be the path homotopy between f and g . Then $F(0, 0) = b_0$. Let $\tilde{F} : I \times I \rightarrow E$ be the lifting of F to E such that $\tilde{F}(0, 0) = e_0$. By the preceding lemma, $\tilde{F}$ is a path homotopy, so that $\tilde{F}(0 \times I) = \{e_0\}$ and $\tilde{F}(1 \times I)$ is a one-point set $\{e_1\}$.

The restriction $\tilde{F}|I \times 0$ of $\tilde{F}$ to the bottom edge of $I \times I$ is a path on E beginning at e_0 that is a lifting of $F|I \times 0$. By uniqueness of path liftings, we must have $\tilde{F}(s, 0) = f(s)$. Similarly, $\tilde{F}|I \times 1$ is a path on E that is a lifting of $F|I \times 1$, and it begins at e_0 because $\tilde{F}(0 \times I) = \{e_0\}$. By uniqueness of path liftings, $\tilde{F}(s, 1) = \tilde{g}(s)$. Therefore, both $\tilde{f}$ and $\tilde{g}$ end at e_1 , and $\tilde{F}$ is a path homotopy between them. ■

Definition. Let $p : E \rightarrow B$ be a covering map; let $b_0 \in B$. Choose e_0 so that $p(e_0) = b_0$. Given an element $[f]$ of $\pi_1(B, b_0)$, let $\tilde{f}$ be the lifting of f to a path in E that begins at e_0 . Let $\phi([f])$ denote the end point $\tilde{f}(1)$ of $\tilde{f}$. Then ϕ is a well-defined set map

$$\phi : \pi_1(B, b_0) \rightarrow p^{-1}(b_0).$$

We call ϕ the *lifting correspondence* derived from the covering map p . It depends of course on the choice of the point e_0 .

Theorem 54.4. Let $p : E \rightarrow B$ be a covering map; let $p(e_0) = b_0$. If E is path connected, then the lifting correspondence

$$\phi : \pi_1(B, b_0) \rightarrow p^{-1}(b_0)$$

is surjective. If E is simply connected, it is bijective.

Proof. If E is path connected, then, given $e_1 \in p^{-1}(b_0)$, there is a path $\tilde{f}$ in E from e_0 to e_1 . Then $f = p \circ \tilde{f}$ is a loop in B at b_0 , and $\phi([f]) = e_1$ by definition.

Suppose E is simply connected. Let $[f]$ and $[g]$ be two elements of $\pi_1(B, b_0)$ such that $\phi([f]) = \phi([g])$. Let $\tilde{f}$ and $\tilde{g}$ be the liftings of f and g , respectively, to paths in E that begin at e_0 , then $\tilde{f}(1) = \tilde{g}(1)$. Since E is simply connected, there is a path homotopy $\tilde{F}$ in E between $\tilde{f}$ and $\tilde{g}$. Then $p \circ \tilde{F}$ is a path homotopy in B between f and g . ■

Theorem 54.5. The fundamental group of S^1 is isomorphic to the additive group of integers.

Proof. Let $p : \mathbb{R} \rightarrow S^1$ be the covering map of Theorem 53.1, let $e_0 = 0$, and let $b_0 = p(e_0)$. Then $p^{-1}(b_0)$ is the set $\mathbb{Z}$ of integers. Since $\mathbb{R}$ is simply connected, the lifting correspondence

$$\phi : \pi_1(S^1, b_0) \rightarrow \mathbb{Z}$$

is bijective. We show that ϕ is a homomorphism, and the theorem is proved.

Given $[f]$ and $[g]$ in $\pi_1(B, b_0)$, let $\tilde{f}$ and $\tilde{g}$ be their respective liftings to paths on $\mathbb{R}$ beginning at 0. Let $n = \tilde{f}(1)$ and $m = \tilde{g}(1)$; then $\phi([f]) = n$ and $\phi([g]) = m$, by definition. Let $\tilde{\tilde{g}}$ be the path

$$\tilde{\tilde{g}}(s) = n + \tilde{g}(s)$$

on $\mathbb{R}$. Because $p(n+x) = p(x)$ for all $x \in \mathbb{R}$, the path $\tilde{\tilde{g}}$ is a lifting of g ; it begins at n . Then the product $\tilde{f} * \tilde{\tilde{g}}$ is defined, and it is the lifting of $f * g$ that begins at 0, as you can check. The end point of this path is $\tilde{\tilde{g}}(1) = n + m$. Then by definition,

$$\phi([f] * [g]) = n + m = \phi([f]) + \phi([g]). \quad \blacksquare$$

Definition. Let G be a group; let x be an element of G . We denote the inverse of x by x^{-1} . The symbol x^n denotes the n -fold product of x with itself, x^{-n} denotes the n -fold product of x^{-1} with itself, and x^0 denotes the identity element of G . If the set of all elements of the form x^m , for $m \in \mathbb{Z}$, equals G , then G is said to be a *cyclic* group, and x is said to be a *generator* of G .

The cardinality of a group is also called the *order* of the group. A group is cyclic of infinite order if and only if it is isomorphic to the additive group of integers; it is cyclic of order k if and only if it is isomorphic to the group $\mathbb{Z}/k$ of integers modulo k . The preceding theorem implies that the fundamental group of the circle is infinite cyclic.

Note that if x is a generator of the infinite cyclic group G , and if y is an element of the arbitrary group H , then there is a unique homomorphism h of G into H such that $h(x) = y$; it is defined by setting $h(x^n) = y^n$ for all n .

For later use, in §65 and in Chapters 13 and 14, we prove here a strengthened version of Theorem 54.4.

***Theorem 54.6.** Let $p : E \rightarrow B$ be a covering map; let $p(e_0) = b_0$.

(a) The homomorphism $p_* : \pi_1(E, e_0) \rightarrow \pi_1(B, b_0)$ is a monomorphism.

(b) Let $H = p_*(\pi_1(E, e_0))$. The lifting correspondence ϕ induces an injective map

$$\Phi : \pi_1(B, b_0)/H \rightarrow p^{-1}(b_0)$$

of the collection of right cosets of H into $p^{-1}(b_0)$, which is bijective if E is path connected.

(c) If f is a loop in B based at b_0 , then $[f] \in H$ if and only if f lifts to a loop in E based at e_0 .

Proof. (a) Suppose $\tilde{h}$ is a loop in E at e_0 , and $p_*([\tilde{h}])$ is the identity element. Let F be a path homotopy between $p \circ \tilde{h}$ and the constant loop. If $\tilde{F}$ is the lifting of F to E such that $\tilde{F}(0, 0) = e_0$, then $\tilde{F}$ is a path homotopy between $\tilde{h}$ and the constant loop at e_0 .

(b) Given loops f and g in B , let $\tilde{f}$ and $\tilde{g}$ be liftings of them to E that begin at e_0 . Then $\phi([f]) = \tilde{f}(1)$ and $\phi([g]) = \tilde{g}(1)$. We show that $\phi([f]) = \phi([g])$ if and only if $[f] \in H * [g]$.

First, suppose that $[f] \in H * [g]$. Then $[f] = [h * g]$, where $h = p \circ \tilde{h}$ for some loop $\tilde{h}$ in E based at e_0 . Now the product $\tilde{h} * \tilde{g}$ is defined, and it is a lifting of $h * g$. Because $[f] = [h * g]$, the liftings $\tilde{f}$ and $\tilde{h} * \tilde{g}$, which begin at e_0 , must end at the same point of E . Then $\tilde{f}$ and $\tilde{g}$ end at the same point of E , so that $\phi([f]) = \phi([g])$. See Figure 54.3.

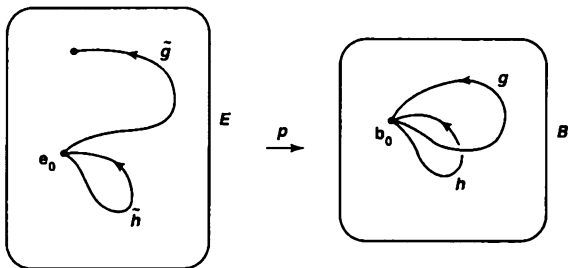


Figure 54.3

Now suppose that $\phi([f]) = \phi([g])$. Then $\tilde{f}$ and $\tilde{g}$ end at the same point of E . The product of $\tilde{f}$ and the reverse of $\tilde{g}$ is defined, and it is a loop $\tilde{h}$ in E based at e_0 . By direct computation, $[\tilde{h} * \tilde{g}] = [\tilde{f}]$. If $\tilde{F}$ is a path homotopy in E between the loops $\tilde{h} * \tilde{g}$ and $\tilde{f}$, then $p \circ \tilde{F}$ is a path homotopy in B between $h * g$ and f , where $h = p \circ \tilde{h}$. Thus $[f] \in H * [g]$, as desired.

If E is path connected, then ϕ is surjective, so that Φ is surjective as well.

(c) Injectivity of Φ means that $\phi([f]) = \phi([g])$ if and only if $[f] \in H * [g]$. Applying this result in the case where g is the constant loop, we see that $\phi([f]) = e_0$ if and only if $[f] \in H$. But $\phi([f]) = e_0$ precisely when the lift of f that begins at e_0 also ends at e_0 . ■

Exercises

1. What goes wrong with the “path-lifting lemma” (Lemma 54.1) for the local homeomorphism of Example 2 of §53?
2. In defining the map $\tilde{F}$ in the proof of Lemma 54.2, why were we so careful about the order in which we considered the small rectangles?
3. Let $p : E \rightarrow B$ be a covering map. Let α and β be paths in B with $\alpha(1) = \beta(0)$; let $\tilde{\alpha}$ and $\tilde{\beta}$ be liftings of them such that $\tilde{\alpha}(1) = \tilde{\beta}(0)$. Show that $\tilde{\alpha} * \tilde{\beta}$ is a lifting of $\alpha * \beta$.

4. Consider the covering map $p : \mathbb{R} \times \mathbb{R}_+ \rightarrow \mathbb{R}^2 - \mathbf{0}$ of Example 6 of §53. Find liftings of the paths

$$f(t) = (2 - t, 0),$$

$$g(t) = ((1 + t) \cos 2\pi t, (1 + t) \sin 2\pi t)$$

$$h(t) = f * g.$$

Sketch these paths and their liftings.

5. Consider the covering map $p \times p : \mathbb{R} \times \mathbb{R} \rightarrow S^1 \times S^1$ of Example 4 of §53. Consider the path

$$f(t) = (\cos 2\pi t, \sin 2\pi t) \times (\cos 4\pi t, \sin 4\pi t)$$

in $S^1 \times S^1$. Sketch what f looks like when $S^1 \times S^1$ is identified with the doughnut surface D . Find a lifting $\tilde{f}$ of f to $\mathbb{R} \times \mathbb{R}$, and sketch it.

6. Consider the maps $g, h : S^1 \rightarrow S^1$ given $g(z) = z^n$ and $h(z) = 1/z^n$. (Here we represent S^1 as the set of complex numbers z of absolute value 1.) Compute the induced homomorphisms g_*, h_* of the infinite cyclic group $\pi_1(S^1, b_0)$ into itself. [Hint: Recall the equation $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$.]
7. Generalize the proof of Theorem 54.5 to show that the fundamental group of the torus is isomorphic to the group $\mathbb{Z} \times \mathbb{Z}$.
8. Let $p : E \rightarrow B$ be a covering map, with E path connected. Show that if B is simply connected, then p is a homeomorphism.

§55 Retractions and Fixed Points

We now prove several classical results of topology that follow from our knowledge of the fundamental group of S^1 .

Definition. If $A \subset X$, a **retraction** of X onto A is a continuous map $r : X \rightarrow A$ such that $r|_A$ is the identity map of A . If such a map r exists, we say that A is a **retract** of X .

Lemma 55.1. *If A is a retract of X , then the homomorphism of fundamental groups induced by inclusion $j : A \rightarrow X$ is injective.*

Proof. If $r : X \rightarrow A$ is a retraction, then the composite map $r \circ j$ equals the identity map of A . It follows that $r_* \circ j_*$ is the identity map of $\pi_1(A, a)$, so that j_* must be injective. ■

Theorem 55.2 (No-retraction theorem). *There is no retraction of B^2 onto S^1 .*

Proof. If S^1 were a retract of B^2 , then the homomorphism induced by inclusion $j : S^1 \rightarrow B^2$ would be injective. But the fundamental group of S^1 is nontrivial and the fundamental group of B^2 is trivial. ■

Lemma 55.3. Let $h : S^1 \rightarrow X$ be a continuous map. Then the following conditions are equivalent:

- (1) h is nullhomotopic.
- (2) h extends to a continuous map $k : B^2 \rightarrow X$.
- (3) h_* is the trivial homomorphism of fundamental groups.

Proof. (1) $\Rightarrow$ (2). Let $H : S^1 \times I \rightarrow X$ be a homotopy between h and a constant map. Let $\pi : S^1 \times I \rightarrow B^2$ be the map

$$\pi(x, t) = (1 - t)x.$$

Then π is continuous, closed and surjective, so it is a quotient map; it collapses $S^1 \times 1$ to the point 0 and is otherwise injective. Because H is constant on $S^1 \times 1$, it induces, via the quotient map π , a continuous map $k : B^2 \rightarrow X$ that is an extension of h . See Figure 55.1.

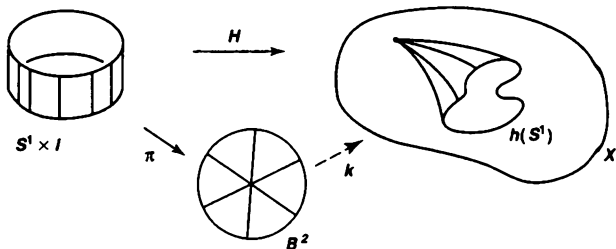


Figure 55.1

(2) $\Rightarrow$ (3). If $j : S^1 \rightarrow B^2$ is the inclusion map, then h equals the composite $k \circ j$. Hence $h_* = k_* \circ j_*$. But

$$j_* : \pi_1(S^1, b_0) \rightarrow \pi_1(B^2, b_0)$$

is trivial because the fundamental group of B^2 is trivial. Therefore h_* is trivial.

(3) $\Rightarrow$ (1). Let $p : \mathbb{R} \rightarrow S^1$ be the standard covering map, and let $p_0 : I \rightarrow S^1$ be its restriction to the unit interval. Then $[p_0]$ generates $\pi_1(S^1, b_0)$ because p_0 is a loop in S^1 whose lift to $\mathbb{R}$ begins at 0 and ends at 1.

Let $x_0 = h(b_0)$. Because h_* is trivial, the loop $f = h \circ p_0$ represents the identity element of $\pi_1(X, x_0)$. Therefore, there is a path homotopy F in X between f and the constant path at x_0 . The map $p_0 \times \text{id} : I \times I \rightarrow S^1 \times I$ is a quotient map, being continuous, closed, and surjective; it maps $0 \times t$ and $1 \times t$ to $b_0 \times t$ for each t , but is otherwise injective. The path homotopy F maps $0 \times I$ and $1 \times I$ and $I \times 1$ to the point x_0 of X , so it induces a continuous map $H : S^1 \times I \rightarrow X$ that is a homotopy between h and a constant map. See Figure 55.2. ■

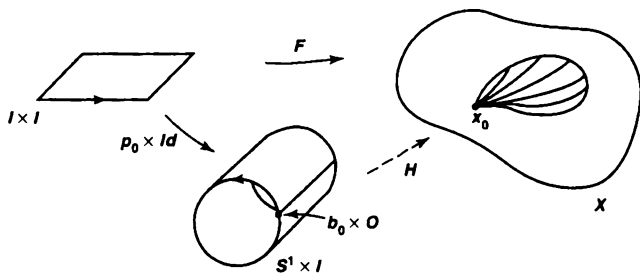


Figure 55.2

Corollary 55.4. The inclusion map $j : S^1 \rightarrow \mathbb{R}^2 - 0$ is not nullhomotopic. The identity map $i : S^1 \rightarrow S^1$ is not nullhomotopic.

Proof. There is a retraction of $\mathbb{R}^2 - 0$ onto S^1 given by the equation $r(x) = x/\|x\|$. Therefore, j_* is injective, and hence nontrivial. Similarly, i_* is the identity homomorphism, and hence nontrivial. ■

Theorem 55.5. Given a nonvanishing vector field on B^2 , there exists a point of S^1 where the vector field points directly inward and a point of S^1 where it points directly outward.

Proof. A **vector field** on B^2 is an ordered pair $(x, v(x))$, where x is in B^2 and v is a continuous map of B^2 into $\mathbb{R}^2$. In calculus, one often uses the notation

$$v(x) = v_1(x)\mathbf{i} + v_2(x)\mathbf{j}$$

for the function v , where $\mathbf{i}$ and $\mathbf{j}$ are the standard unit basis vectors in $\mathbb{R}^2$. But we shall stick with simple functional notation. To say that a vector field is *nonvanishing* means that $v(x) \neq 0$ for every x ; in such a case v actually maps B^2 into $\mathbb{R}^2 - 0$.

We suppose first that $v(x)$ does not point directly inward at any point x of S^1 and derive a contradiction. Consider the map $v : B^2 \rightarrow \mathbb{R}^2 - 0$; let w be its restriction to S^1 . Because the map w extends to a map of B^2 into $\mathbb{R}^2 - 0$, it is nullhomotopic.

On the other hand, w is homotopic to the inclusion map $j : S^1 \rightarrow \mathbb{R}^2 - 0$. Figure 55.3 illustrates the homotopy; one defines it formally by the equation

$$F(x, t) = tx + (1 - t)w(x),$$

for $x \in S^1$. We must show that $F(x, t) \neq 0$. Clearly, $F(x, t) \neq 0$ for $t = 0$ and $t = 1$. If $F(x, t) = 0$ for some t with $0 < t < 1$, then $tx + (1 - t)w(x) = 0$, so that $w(x)$ equals a negative scalar multiple of x . But this means that $w(x)$ points directly inward at x ! Hence F maps $S^1 \times I$ into $\mathbb{R}^2 - 0$, as desired.

It follows that j is nullhomotopic, contradicting the preceding corollary.

To show that v points directly outward at some point of S^1 , we apply the result just proved to the vector field $(x, -v(x))$. ■

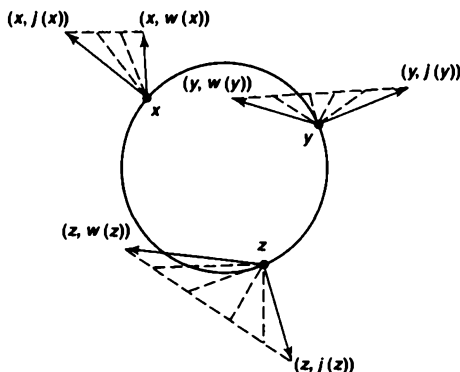


Figure 55.3

We have already seen that every continuous map $f : [0, 1] \rightarrow [0, 1]$ has a fixed point (see Exercise 3 of §24). The same is true for the ball B^2 , although the proof is deeper:

Theorem 55.6 (Brouwer fixed-point theorem for the disc). *If $f : B^2 \rightarrow B^2$ is continuous, then there exists a point $x \in B^2$ such that $f(x) = x$.*

Proof. We proceed by contradiction. Suppose that $f(x) \neq x$ for every x in B^2 . Then defining $v(x) = f(x) - x$ gives us a nonvanishing vector field $(x, v(x))$ on B^2 . But the vector field v cannot point directly outward at any point x of S^1 , for that would mean

$$f(x) - x = ax$$

for some *positive* real number a , so that $f(x) = (1 + a)x$ would lie outside the unit ball B^2 . We thus arrive at a contradiction. ■

One might well wonder why fixed-point theorems are of interest in mathematics. It turns out that many problems, such as problems concerning existence of solutions for systems of equations, for instance, can be formulated as fixed-point problems. Here is one example, a classical theorem of Frobenius. We assume some knowledge of linear algebra at this point.

***Corollary 55.7.** *Let A be a 3 by 3 matrix of positive real numbers. Then A has a positive real eigenvalue (characteristic value).*

Proof. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear transformation whose matrix (relative to the standard basis for $\mathbb{R}^3$) is A . Let B be the intersection of the 2-sphere S^2 with the first

octant

$$\{(x_1, x_2, x_3) \mid x_1 \geq 0 \text{ and } x_2 \geq 0 \text{ and } x_3 \geq 0\}$$

of $\mathbb{R}^3$. It is easy to show that B is homeomorphic to the ball B^2 , so that the fixed-point theorem holds for continuous maps of B into itself.

Now if $x = (x_1, x_2, x_3)$ is in B , then all the components of x are nonnegative and at least one is positive. Because all entries of A are positive, the vector $T(x)$ is a vector all of whose components are positive. As a result, the map $x \rightarrow T(x)/\|T(x)\|$ is a continuous map of B to itself, which therefore has a fixed point x_0 . Then

$$T(x_0) = \|T(x_0)\|x_0,$$

so that T (and therefore the matrix A) has the positive real eigenvalue $\|T(x_0)\|$. ■

Finally, we prove a theorem that implies that the triangular region

$$T = \{(x, y) \mid x \geq 0 \text{ and } y \geq 0 \text{ and } x + y \leq 1\}$$

in $\mathbb{R}^2$ has topological dimension at least 2. (See §50.)

Theorem 55.8. *There is an $\epsilon > 0$ such that for every open covering $\mathcal{A}$ of T by sets of diameter less than ϵ , some point of T belongs to at least three elements of $\mathcal{A}$.*

Proof. We use the fact that T is homeomorphic to B^2 , so that we can apply the results proved in this section to the space T .

Choose $\epsilon > 0$ so that no set of diameter less than ϵ intersects all three edges of T . (In fact, $\epsilon = \frac{1}{2}$ will do.) We suppose that $\mathcal{A} = \{U_1, \dots, U_n\}$ is an open covering of T by sets of diameter less than ϵ , such that no three elements of $\mathcal{A}$ intersect, and derive a contradiction.

For each $i = 1, \dots, n$, choose a vertex v_i of T as follows: If U_i intersects two edges of T , let v_i be the vertex common to these edges. If U_i intersects only one edge of T , let v_i be one of the end points of this edge. If U_i intersects no edge of T , let v_i be any vertex of T .

Now let $\{\phi_i\}$ be a partition of unity dominated by $\{U_1, \dots, U_n\}$. (See §36.) Define $k : T \rightarrow \mathbb{R}^2$ by the equation

$$k(x) = \sum_{i=1}^n \phi_i(x)v_i.$$

Then k is continuous. Given a point x of T , it lies in at most two elements of $\mathcal{A}$; hence at most two of the numbers $\phi_i(x)$ are nonzero. Then $k(x) = v_i$ if x lies in only one open set U_i , and $k(x) = tv_i + (1-t)v_j$ for some t with $0 \leq t \leq 1$ if x lies in two open sets U_i and U_j . In either case, $k(x)$ belongs to the union of the edges of T , which is $\text{Bd } T$. Thus k maps T into $\text{Bd } T$.

Furthermore, k maps each edge of T into itself. For if x belongs to the edge vw of T , any open set U_i containing x intersects this edge, so that v_i must equal either v or w . The definition of k then shows that $k(x)$ belongs to vw .

Let $h : \text{Bd } T \rightarrow \text{Bd } T$ be the restriction of k to $\text{Bd } T$. Since h can be extended to the continuous map k , it is nulhomotopic. On the other hand, h is homotopic to the identity map of $\text{Bd } T$ to itself; indeed, since h maps each edge of T into itself, the straight-line homotopy between h and the identity map of $\text{Bd } T$ is such a homotopy. But the identity map i of $\text{Bd } T$ is *not* nulhomotopic. ■

Exercises

1. Show that if A is a retract of B^2 , then every continuous map $f : A \rightarrow A$ has a fixed point.
2. Show that if $h : S^1 \rightarrow S^1$ is nulhomotopic, then h has a fixed point and h maps some point x to its antipode $-x$.
3. Show that if A is a nonsingular 3 by 3 matrix having nonnegative entries, then A has a positive real eigenvalue.
4. Suppose that you are given the fact that for each n , there is no retraction $r : B^{n+1} \rightarrow S^n$. (This result can be proved using more advanced techniques of algebraic topology.) Prove the following:
 - (a) The identity map $i : S^n \rightarrow S^n$ is not nulhomotopic.
 - (b) The inclusion map $j : S^n \rightarrow \mathbb{R}^{n+1} - \mathbf{0}$ is not nulhomotopic.
 - (c) Every nonvanishing vector field on B^{n+1} points directly outward at some point of S^n , and directly inward at some point of S^n .
 - (d) Every continuous map $f : B^{n+1} \rightarrow B^{n+1}$ has a fixed point.
 - (e) Every $n + 1$ by $n + 1$ matrix with positive real entries has a positive eigenvalue.
 - (f) If $h : S^n \rightarrow S^n$ is nulhomotopic, then h has a fixed point and h maps some point x to its antipode $-x$.

*§56 The Fundamental Theorem of Algebra

It is a basic fact about the complex numbers that every polynomial equation

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0$$

of degree n with real or complex coefficients has n roots (if the roots are counted according to their multiplicities). You probably first were told this fact in high school algebra, although it is doubtful that it was proved for you at that time.

The proof is, in fact, rather hard; the most difficult part is to prove that every polynomial equation of positive degree has *at least one* root. There are various ways

of doing this. One can use only techniques of algebra; this proof is long and arduous. Or one can develop the theory of analytic functions of a complex variable to the point where it becomes a trivial corollary of Liouville's theorem. Or one can prove it as a relatively easy corollary of our computation of the fundamental group of the circle; this we do now.

Theorem 56.1 (The fundamental theorem of algebra). *A polynomial equation*

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0$$

of degree $n > 0$ with real or complex coefficients has at least one (real or complex) root.

Proof. *Step 1.* Consider the map $f : S^1 \rightarrow S^1$ given by $f(z) = z^n$, where z is a complex number. We show that the induced homomorphism f_* of fundamental groups is injective.

Let $p_0 : I \rightarrow S^1$ be the standard loop in S^1 ,

$$p_0(s) = e^{2\pi is} = (\cos 2\pi s, \sin 2\pi s).$$

Its image under f_* is the loop

$$f(p_0(s)) = (e^{2\pi is})^n = (\cos 2\pi ns, \sin 2\pi ns).$$

This loop lifts to the path $s \rightarrow ns$ in the covering space $\mathbb{R}$. Therefore, the loop $f \circ p_0$ corresponds to the integer n under the standard isomorphism of $\pi_1(S^1, b_0)$ with the integers, whereas p_0 corresponds to the number 1. Thus f_* is "multiplication by n " in the fundamental group of S^1 , so that in particular, f_* is injective.

Step 2. We show that if $g : S^1 \rightarrow \mathbb{R}^2 - \mathbf{0}$ is the map $g(z) = z^n$, then g is not nulhomotopic.

The map g equals the map f of Step 1 followed by the inclusion map $j : S^1 \rightarrow \mathbb{R}^2 - \mathbf{0}$. Now f_* is injective, and j_* is injective because S^1 is a retract of $\mathbb{R}^2 - \mathbf{0}$. Therefore, $g_* = j_* \circ f_*$ is injective. Thus g cannot be nulhomotopic.

Step 3. Now we prove a special case of the theorem. Given a polynomial equation

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0,$$

we assume that

$$|a_{n-1}| + \cdots + |a_1| + |a_0| < 1$$

and show that the equation has a root lying in the unit ball B^2 .

Assume it has no such root. Then we can define a map $k : B^2 \rightarrow \mathbb{R}^2 - \mathbf{0}$ by the equation

$$k(z) = z^n + a_{n-1}z^{n-1} + \cdots + a_1z + a_0.$$

Let h be the restriction of k to S^1 . Because h extends to a map of the unit ball into $\mathbb{R}^2 - 0$, the map h is nulhomotopic.

On the other hand, we shall define a homotopy F between h and the map g of Step 2; since g is not nulhomotopic, we have a contradiction. We define $F : S^1 \times I \rightarrow \mathbb{R}^2 - 0$ by the equation

$$F(z, t) = z^n + t(a_{n-1}z^{n-1} + \cdots + a_0).$$

See Figure 56.1; $F(z, t)$ never equals 0 because

$$\begin{aligned} |F(z, t)| &\geq |z^n| - |t(a_{n-1}z^{n-1} + \cdots + a_0)| \\ &\geq 1 - t(|a_{n-1}z^{n-1}| + \cdots + |a_0|) \\ &= 1 - t(|a_{n-1}| + \cdots + |a_0|) > 0. \end{aligned}$$

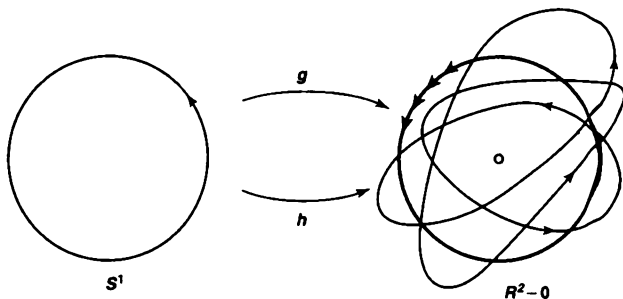


Figure 56.1

Step 4. Now we prove the general case. Given a polynomial equation

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0,$$

let us choose a real number $c > 0$ and substitute $x = cy$. We obtain the equation

$$(cy)^n + a_{n-1}(cy)^{n-1} + \cdots + a_1(cy) + a_0 = 0$$

or

$$y^n + \frac{a_{n-1}}{c}y^{n-1} + \cdots + \frac{a_1}{c^{n-1}}y + \frac{a_0}{c^n} = 0.$$

If this equation has the root $y = y_0$, then the original equation has the root $x_0 = cy_0$. So we need merely choose c large enough that

$$\left| \frac{a_{n-1}}{c} \right| + \left| \frac{a_{n-2}}{c^2} \right| + \cdots + \left| \frac{a_1}{c^{n-1}} \right| + \left| \frac{a_0}{c^n} \right| < 1$$

to reduce the theorem to the special case considered in Step 3. ■

Exercises

1. Given a polynomial equation

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = 0$$

with real or complex coefficients. Show that if $|a_{n-1}| + \cdots + |a_1| + |a_0| < 1$, then *all* the roots of the equation lie interior to the unit ball B^2 . [Hint: Let $g(x) = 1 + a_{n-1}x + \cdots + a_1x^{n-1} + a_0x^n$, and show that $g(x) \neq 0$ for $x \in B^2$.]

2. Find a circle about the origin containing all the roots of the polynomial equation $x^7 + x^2 + 1 = 0$.

*§57 The Borsuk-Ulam Theorem

Here is a “brain-teaser” problem: Suppose you are given a bounded polygonal region A in the plane $\mathbb{R}^2$. No matter what shape A has, it is easy to show that there exists a straight line that bisects A , that is, one that cuts the area of A in half. Simply take the horizontal line $y = c$, let $f(c)$ denote the area of that part of A that lies beneath this line, note that f is a continuous function of c , and use the intermediate-value theorem to find a value of c for which $f(c)$ equals exactly half the area of A .

But now suppose instead that you are given *two* such regions A_1 and A_2 , you are asked to find a single line that bisects them *both*. It is not obvious even that there exists such a line. Try to find one for an arbitrary pair of triangular regions if you have doubts!

In fact, such a line always exists. This result is a corollary of a well-known theorem called the Borsuk-Ulam theorem, to which we now turn.

Definition. If x is a point of S^n , then its *antipode* is the point $-x$. A map $h : S^n \rightarrow S^m$ is said to be *antipode-preserving* if $h(-x) = -h(x)$ for all $x \in S^n$.

Theorem 57.1. If $h : S^1 \rightarrow S^1$ is continuous and antipode-preserving, then h is not nullhomotopic.

Proof. Let b_0 be the point $(1, 0)$ of S^1 . Let $\rho : S^1 \rightarrow S^1$ be a rotation of S^1 that maps $h(b_0)$ to b_0 . Since ρ preserves antipodes, so does the composite $\rho \circ h$. Furthermore, if H were a homotopy between h and a constant map, then $\rho \circ H$ would be a homotopy between $\rho \circ h$ and a constant map. Therefore, it suffices to prove the theorem under the additional hypothesis that $h(b_0) = b_0$.

Step 1. Let $q : S^1 \rightarrow S^1$ be the map $q(z) = z^2$, where z is a complex number. Or in real coordinates, $q(\cos \theta, \sin \theta) = (\cos 2\theta, \sin 2\theta)$. The map q is a quotient map, being continuous, closed, and surjective. The inverse image under q of any point of S^1 consists of two antipodal points z and $-z$ of S^1 . Because $h(-z) = -h(z)$, one has the

equation $q(h(-z)) = q(h(z))$. Therefore, because q is a quotient map, the map $q \circ h$ induces a continuous map $k : S^1 \rightarrow S^1$ such that $k \circ q = q \circ h$.

$$\begin{array}{ccc} S^1 & \xrightarrow{h} & S^1 \\ q \downarrow & & \downarrow q \\ S^1 & \xrightarrow{k} & S^1 \end{array}$$

Note that $q(b_0) = h(b_0) = b_0$, so that $k(b_0) = b_0$ as well. Also, $h(-b_0) = -b_0$.

Step 2. We show that the homomorphism k_* of $\pi_1(S^1, b_0)$ with itself is nontrivial.

For this purpose, we first show that q is a covering map. (We gave this as an exercise in §53.) The proof is similar to the proof that the standard map $p : \mathbb{R} \rightarrow S^1$ is a covering map. If, for instance, U is the subset of S^1 consisting of those points having positive second coordinate, then $p^{-1}(U)$ consist of those points of S^1 lying in the first and third quadrants of $\mathbb{R}^2$. The map q carries each of these sets homeomorphically onto U . Similar arguments apply when U is the intersection of S^1 with the open lower half-plane, or with the open right and left half-planes.

Second, we note that if $\tilde{f}$ is any path in S^1 from b_0 to $-b_0$, then the loop $f = q \circ \tilde{f}$ represents a nontrivial element of $\pi_1(S^1, b_0)$. For $\tilde{f}$ is a lifting of f to S^1 that begins at b_0 and does not end at b_0 .

Finally, we show k_* is nontrivial. Let $\tilde{f}$ be a path in S^1 from b_0 to $-b_0$, and let f be the loop $q \circ \tilde{f}$. Then $k_*[f]$ is not trivial, for $k_*[f] = [k \circ (q \circ \tilde{f})] = [q \circ (h \circ \tilde{f})]$; the latter is nontrivial because $h \circ \tilde{f}$ is a path in S^1 from b_0 to $-b_0$.

Step 3. Finally, we show that the homomorphism h_* is nontrivial, so that h cannot be nulhomotopic.

The homomorphism k_* is injective, being a nontrivial homomorphism of an infinite cyclic group with itself. The homomorphism q_* is also injective; indeed, q_* corresponds to multiplication by two in the group of integers. It follows that $k_* \circ q_*$ is injective. Since $q_* \circ h_* = k_* \circ q_*$, the homomorphism h_* must be injective as well. ■

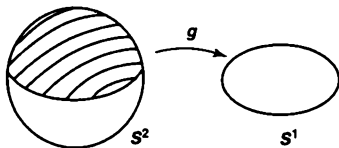


Figure 57.1

Theorem 57.2. *There is no continuous antipode-preserving map $g : S^2 \rightarrow S^1$.*

Proof. Suppose $g : S^2 \rightarrow S^1$ is continuous and antipode preserving. Let us take S^1 to be the equator of S^2 . Then the restriction of g to S^1 is a continuous antipode-preserving map h of S^1 to itself. By the preceding theorem, h is not nulhomotopic. But the upper hemisphere E of S^2 is homeomorphic to the ball B^2 , and g is a continuous extension of h to E ! See Figure 57.1. ■

Theorem 57.3 (Borsuk-Ulam theorem for S^2). Given a continuous map $f : S^2 \rightarrow \mathbb{R}^2$, there is a point x of S^2 such that $f(x) = f(-x)$.

Proof. Suppose that $f(x) \neq f(-x)$ for all $x \in S^2$. Then the map

$$g(x) = [f(x) - f(-x)] / \|f(x) - f(-x)\|$$

is a continuous map $g : S^2 \rightarrow S^1$ such that $g(-x) = -g(x)$ for all x . ■

Theorem 57.4 (The bisection theorem). Given two bounded polygonal regions in $\mathbb{R}^2$, there exists a line in $\mathbb{R}^2$ that bisects each of them.

Proof. We take two bounded polygonal regions A_1 and A_2 in the plane $\mathbb{R}^2 \times 1$ in $\mathbb{R}^3$, and show there is a line L in this plane that bisects each of them.

Given a point u of S^2 , let us consider the plane P in $\mathbb{R}^3$ passing through the origin that has u as its unit normal vector. This plane divides $\mathbb{R}^3$ into two half-spaces; let $f_i(u)$ equal the area of that portion of A_i that lies on the same side of P as does the vector u .

If u is the unit vector $\mathbf{k}$, then $f_i(u) = \text{area } A_i$; and if $u = -\mathbf{k}$, then $f_i(u) = 0$. Otherwise, the plane P intersects the plane $\mathbb{R}^2 \times 1$ in a line L that splits $\mathbb{R}^2 \times 1$ into two half-planes, and $f_i(u)$ is the area of that part of A_i that lies on one side of this line. See Figure 57.2.

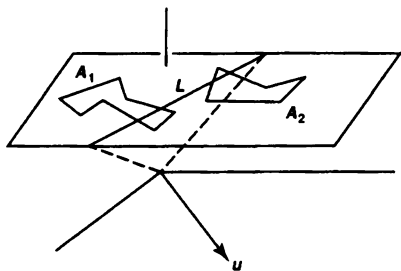


Figure 57.2

Replacing u by $-u$ gives us the same plane P , but the other half-space, so that $f_i(-u)$ is the area of that part of A_i that lies on the other side of P from u . It follows that

$$f_i(u) + f_i(-u) = \text{area } A_i.$$

Now consider the map $F : S^2 \rightarrow \mathbb{R}^2$ given by $F(u) = (f_1(u), f_2(u))$. The Borsuk-Ulam theorem gives us a point u of S^2 for which $F(u) = F(-u)$. Then $f_i(u) = f_i(-u)$ for $i = 1, 2$, that $f_i(u) = \frac{1}{2} \text{area } A_i$, as desired. ■

We have proved the bisection theorem for bounded polygonal regions in the plane. However, all that was needed in the proof was the existence of an additive area function for A_1 and A_2 . Thus, the theorem holds for any two sets A_1 and A_2 that are “Jordan-measurable” in the sense used in analysis.

These theorems generalize to higher dimensions, but the proofs are considerably more sophisticated. The generalized version of the bisection theorem states that given n Jordan-measurable sets in $\mathbb{R}^n$, there exists a plane of dimension $n - 1$ that bisects them all. In the case $n = 3$, this result goes by the pleasant name of the “ham sandwich theorem.” If one considers a ham sandwich to consist of two pieces of bread and a slab of ham, then the bisection theorem says that one can divide each of them precisely in half with a single whack of a cleaver!

Exercises

1. Prove the following “theorem of meteorology”: At any given moment in time, there exists a pair of antipodal points on the surface of the earth at which both the temperature and the barometric pressure are equal.
2. Show that if $g : S^2 \rightarrow S^2$ is continuous and $g(x) \neq g(-x)$ for all x , then g is surjective. [Hint: If $p \in S^2$, then $S^2 - \{p\}$ is homeomorphic to $\mathbb{R}^2$.]
3. Let $h : S^1 \rightarrow S^1$ be continuous and antipode-preserving with $h(b_0) = b_0$. Show that h_* carries a generator of $\pi_1(S^1, b_0)$ to an odd power of itself. [Hint: If k is the map constructed in the proof of Theorem 57.1, show that k_* does the same.]
4. Suppose you are given the fact that for each n , no continuous antipode-preserving map $h : S^n \rightarrow S^n$ is nullhomotopic. (This result can be proved using more advanced techniques of algebraic topology.) Prove the following:
 - (a) There is no retraction $r : B^{n+1} \rightarrow S^n$.
 - (b) There is no continuous antipode-preserving map $g : S^{n+1} \rightarrow S^n$.
 - (c) (Borsuk-Ulam theorem) Given a continuous map $f : S^{n+1} \rightarrow \mathbb{R}^{n+1}$, there is a point x of S^{n+1} such that $f(x) = f(-x)$.
 - (d) If $A_1, \dots, A_{n+1}$ are bounded measurable sets in $\mathbb{R}^{n+1}$, there exists an n -plane in $\mathbb{R}^{n+1}$ that bisects each of them.

§58 Deformation Retracts and Homotopy Type

As we have seen, one way of obtaining information about the fundamental group of a space X is to study the covering spaces of X . Another is one we discuss in this section, which involves the notion of *homotopy type*. It provides a method for reducing the problem of computing the fundamental group of a space to that of computing the fundamental group of some other space—preferably, one that is more familiar.

We begin with a lemma.

Lemma 58.1. Let $h, k : (X, x_0) \rightarrow (Y, y_0)$ be continuous maps. If h and k are homotopic, and if the image of the base point x_0 of X remains fixed at y_0 during the homotopy, then the homomorphisms h_* and k_* are equal.

Proof. The proof is immediate. By assumption, there is a homotopy $H : X \times I \rightarrow Y$ between h and k such that $H(x_0, t) = y_0$ for all t . It follows that if f is a loop in X based at x_0 , then the composite

$$I \times I \xrightarrow{f \times \text{id}} X \times I \xrightarrow{H} Y$$

is a homotopy between $h \circ f$ and $k \circ f$; it is a path homotopy because f is a loop at x_0 and H maps $x_0 \times I$ to y_0 . ■

Using this lemma, we generalize a result about the space $\mathbb{R}^2 - \mathbf{0}$ proved earlier, proving that the homomorphism induced by inclusion $j : S^1 \rightarrow \mathbb{R}^2 - \mathbf{0}$ is not only injective but surjective as well. More generally, we prove the following:

Theorem 58.2. The inclusion map $j : S^n \rightarrow \mathbb{R}^{n+1} - \mathbf{0}$ induces an isomorphism of fundamental groups.

Proof. Let $X = \mathbb{R}^{n+1} - \mathbf{0}$; let $b_0 = (1, 0, \dots, 0)$. Let $r : X \rightarrow S^n$ be the map $r(x) = x/\|x\|$. Then $r \circ j$ is the identity map of S^n , so that $r_* \circ j_*$ is the identity homomorphism of $\pi_1(S^n, b_0)$.

Now consider the composite $j \circ r$, which maps X to itself;

$$X \xrightarrow{r} S^n \xrightarrow{j} X.$$

This map is not the identity map of X , but it is homotopic to the identity map. Indeed, the straight-line homotopy $H : X \times I \rightarrow X$, given by

$$H(x, t) = (1 - t)x + tx/\|x\|,$$

is a homotopy between the identity map of X and the map $j \circ r$. For $H(x, t)$ is never equal to $\mathbf{0}$, because $(1 - t) + t/\|x\|$ is a number between 1 and $1/\|x\|$. Note that the point b_0 remains fixed during the homotopy, since $\|b_0\| = 1$. It follows from the preceding lemma that the homomorphism $(j \circ r)_* = j_* \circ r_*$ is the identity homomorphism of $\pi_1(X, b_0)$. ■

What made the preceding proof work? Roughly speaking, it worked because we had a natural way of deforming the identity map of $\mathbb{R}^{n+1} - \mathbf{0}$ to a map that collapsed all of $\mathbb{R}^{n+1} - \mathbf{0}$ onto S^n . The deformation H gradually collapsed each radial line emanating from the origin to the point where it intersected S^n ; each point of S^n remained fixed during this deformation.

Figure 58.1 illustrates, in the case $n = 1$, how the deformation H gives rise to a path homotopy $H(f(s), t)$ between the loop f in $\mathbb{R}^2 - \mathbf{0}$ and the loop $g = f/\|f\|$ in S^1 .

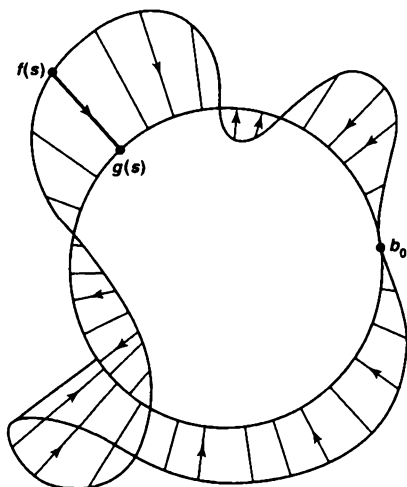


Figure 58.1

These comments lead us to formulate a more general situation in which the same procedure applies.

Definition. Let A be a subspace of X . We say that A is a **deformation retract** of X if the identity map of X is homotopic to a map that carries all of X into A , such that each point of A remains fixed during the homotopy. This means that there is a continuous map $H : X \times I \rightarrow X$ such that $H(x, 0) = x$ and $H(x, 1) \in A$ for all $x \in X$, and $H(a, t) = a$ for all $a \in A$. The homotopy H is called a **deformation retraction** of X onto A . The map $r : X \rightarrow A$ defined by the equation $r(x) = H(x, 1)$ is a retraction of X onto A , and H is a homotopy between the identity map of X and the map $j \circ r$, where $j : A \rightarrow X$ is inclusion.

The proof of the preceding theorem generalizes immediately to prove the following:

Theorem 58.3. Let A be a deformation retract of X ; let $x_0 \in A$. Then the inclusion map

$$j : (A, x_0) \rightarrow (X, x_0)$$

induces an isomorphism of fundamental groups.

EXAMPLE 1. Let B denote the z -axis in $\mathbb{R}^3$. Consider the space $\mathbb{R}^3 - B$. It has the punctured xy -plane $(\mathbb{R}^2 - 0) \times 0$ as a deformation retract. The map H defined by the equation

$$H(x, y, z, t) = (x, y, (1-t)z)$$

is a deformation retraction; it gradually collapses each line parallel to the z -axis into the point where the line intersects the xy -plane. We conclude that the space $\mathbb{R}^3 - B$ has an infinite cyclic fundamental group.

EXAMPLE 2. Consider $\mathbb{R}^2 - p - q$, the *doubly punctured plane*. We assert it has the "figure eight" space as a deformation retract. Rather than writing equations, we merely sketch the deformation retraction; it is the three-stage deformation indicated in Figure 58.2.

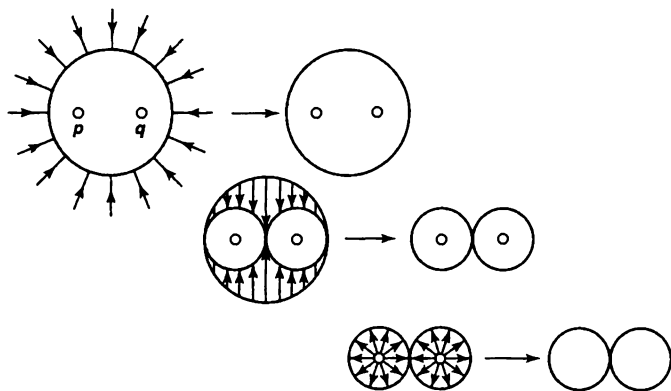


Figure 58.2

EXAMPLE 3. Another deformation retract of $\mathbb{R}^2 - p - q$ is the "theta space"

$$\theta = S^1 \cup (0 \times [-1, 1]);$$

we leave it to you to sketch the maps involved. As a result, the figure eight and the theta space have isomorphic fundamental groups, even though neither is a deformation retract of the other.

Of course, we do not know anything about the fundamental group of the figure eight as yet. But we shall.

The example of the figure eight and the theta space suggests the possibility that there might be a more general way of showing two spaces have isomorphic fundamental groups than by showing that one is homeomorphic to a deformation retract of the other. We formulate such a notion now.

Definition. Let $f : X \rightarrow Y$ and $g : Y \rightarrow X$ be continuous maps. Suppose that the map $g \circ f : X \rightarrow X$ is homotopic to the identity map of X , and the map $f \circ g : Y \rightarrow Y$ is homotopic to the identity map of Y . Then the maps f and g are called **homotopy equivalences**, and each is said to be a **homotopy inverse** of the other.

It is straightforward to show that if $f : X \rightarrow Y$ is a homotopy equivalence of X with Y and $h : Y \rightarrow Z$ is a homotopy equivalence of Y with Z , then $h \circ f : X \rightarrow Z$ is a homotopy equivalence of X with Z . It follows that the relation of homotopy equivalence is an equivalence relation. Two spaces that are homotopy equivalent are said to have the same **homotopy type**.

Note that if A is a deformation retract of X , then A has the same homotopy type as X . For let $j : A \rightarrow X$ be the inclusion mapping and let $r : X \rightarrow A$ be the retraction mapping. Then the composite $r \circ j$ equals the identity map of A , and the composite $j \circ r$ is by hypothesis homotopic to the identity map of X (and in fact each point of A remains fixed during the homotopy).

We now show that two spaces having the same homotopy type have isomorphic fundamental groups. For this purpose, we need to study what happens when we have a homotopy between two continuous maps of X into Y such that the base point of X does *not* remain fixed during the homotopy.

Lemma 58.4. Let $h, k : X \rightarrow Y$ be continuous maps; let $h(x_0) = y_0$ and $k(x_0) = y_1$. If h and k are homotopic, there is a path α in Y from y_0 to y_1 such that $k_* = \hat{\alpha} \circ h_*$. Indeed, if $H : X \times I \rightarrow Y$ is the homotopy between h and k , then α is the path $\alpha(t) = H(x_0, t)$.

$$\begin{array}{ccc} \pi_1(X, x_0) & \xrightarrow{h_*} & \pi_1(Y, y_0) \\ & \searrow k_* & \downarrow \hat{\alpha} \\ & & \pi_1(Y, y_1) \end{array}$$

Proof. Let $f : I \rightarrow X$ be a loop in X based at x_0 . We must show that

$$k_*([f]) = \hat{\alpha}(h_*([f])).$$

This equation states that $[k \circ f] = [\tilde{\alpha}] * [h \circ f] * [\alpha]$, or equivalently, that

$$[\alpha] * [k \circ f] = [h \circ f] * [\alpha].$$

This is the equation we shall verify.

To begin, consider the loops f_0 and f_1 in the space $X \times I$ given by the equations

$$f_0(s) = (f(s), 0) \quad \text{and} \quad f_1(s) = (f(s), 1).$$

Consider also the path c in $X \times I$ given by the equation

$$c(t) = (x_0, t).$$

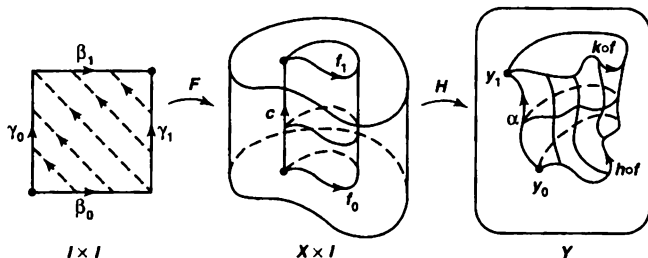


Figure 58.3

Then $H \circ f_0 = h \circ f$ and $H \circ f_1 = k \circ f$, while $H \circ c$ equals the path α . See Figure 58.3.

Let $F : I \times I \rightarrow X \times I$ be the map $F(s, t) = (f(s), t)$. Consider the following paths in $I \times I$, which run along the four edges of $I \times I$:

$$\begin{aligned} \beta_0(s) &= (s, 0) & \text{and} & & \beta_1(s) &= (s, 1), \\ \gamma_0(t) &= (0, t) & \text{and} & & \gamma_1(t) &= (1, t). \end{aligned}$$

Then $F \circ \beta_0 = f_0$ and $F \circ \beta_1 = f_1$, while $F \circ \gamma_0 = F \circ \gamma_1 = c$.

The broken-line paths $\beta_0 * \gamma_1$ and $\gamma_0 * \beta_1$ are paths in $I \times I$ from $(0, 0)$ to $(1, 1)$; since $I \times I$ is convex, there is a path homotopy G between them. Then $F \circ G$ is a path homotopy in $X \times I$ between $f_0 * c$ and $c * f_1$. And $H \circ (F \circ G)$ is a path homotopy in Y between

$$\begin{aligned} (H \circ f_0) * (H \circ c) &= (h \circ f) * \alpha & \text{and} \\ (H \circ c) * (H \circ f_1) &= \alpha * (k \circ f), \end{aligned}$$

as desired. ■

Corollary 58.5. Let $h, k : X \rightarrow Y$ be homotopic continuous maps; let $h(x_0) = y_0$ and $k(x_0) = y_1$. If h_* is injective, or surjective, or trivial, so is k_* .

Corollary 58.6. Let $h : X \rightarrow Y$. If h is nullhomotopic, then h_* is the trivial homomorphism.

Proof. The constant map induces the trivial homomorphism. ■

Theorem 58.7. Let $f : X \rightarrow Y$ be continuous; let $f(x_0) = y_0$. If f is a homotopy equivalence, then

$$f_* : \pi_1(X, x_0) \longrightarrow \pi_1(Y, y_0)$$

is an isomorphism.

Proof. Let $g : Y \rightarrow X$ be a homotopy inverse for f . Consider the maps

$$(X, x_0) \xrightarrow{f} (Y, y_0) \xrightarrow{g} (X, x_1) \xrightarrow{f} (Y, y_1),$$

where $x_1 = g(y_0)$ and $y_1 = f(x_1)$. We have the corresponding induced homomorphisms:

$$\begin{array}{ccc} \pi_1(X, x_0) & \xrightarrow{(f_{x_0})_*} & \pi_1(Y, y_0) \\ & \searrow g_* & \\ \pi_1(X, x_1) & \xrightarrow{(f_{x_1})_*} & \pi_1(Y, y_1) \end{array}$$

[Here we have to distinguish between the homomorphisms induced by f relative to two different base points.] Now

$$g \circ f : (X, x_0) \longrightarrow (X, x_1)$$

is by hypothesis homotopic to the identity map, so there is a path α in X such that

$$(g \circ f)_* = \hat{\alpha} \circ (i_X)_* = \hat{\alpha}.$$

It follows that $(g \circ f)_* = g_* \circ (f_{x_0})_*$ is an isomorphism.

Similarly, because $f \circ g$ is homotopic to the identity map i_Y , the homomorphism $(f \circ g)_* = (f_{x_1})_* \circ g_*$ is an isomorphism.

The first fact implies that g_* is surjective, and the second implies that g_* is injective. Therefore, g_* is an isomorphism. Applying the first equation once again, we conclude that

$$(f_{x_0})_* = (g_*)^{-1} \circ \hat{\alpha},$$

so that $(f_{x_0})_*$ is also an isomorphism.

Note that although g is a homotopy inverse for f , the homomorphism g_* is not an inverse for the homomorphism $(f_{x_0})_*$. ■

The relation of homotopy equivalence is clearly more general than the notion of deformation retraction. The theta space and the figure eight are both deformation retracts of the doubly punctured plane. Therefore, they are homotopy equivalent to the doubly punctured plane, and hence to each other. But neither is homeomorphic to a deformation retract of the other; in fact, neither of them can even be imbedded in the other.

It is a striking fact that the situation that occurs for these two spaces is the standard situation regarding homotopy equivalences. Martin Fuchs has proved a theorem to the effect that two spaces X and Y have the same homotopy type if and only if they are homeomorphic to deformation retracts of a single space Z . The proof, although it uses only elementary tools, is difficult [F].

Exercises

1. Show that if A is a deformation retract of X , and B is a deformation retract of A , then B is a deformation retract of X .
2. For each of the following spaces, the fundamental group is either trivial, infinite cyclic, or isomorphic to the fundamental group of the figure eight. Determine for each space which of the three alternatives holds.
 - (a) The "solid torus," $B^2 \times S^1$.
 - (b) The torus T with a point removed.
 - (c) The cylinder $S^1 \times I$.
 - (d) The infinite cylinder $S^1 \times \mathbb{R}$.
 - (e) $\mathbb{R}^3$ with the nonnegative x , y , and z axes deleted.

The following subsets of $\mathbb{R}^2$:

- (f) $\{x \mid \|x\| > 1\}$
 - (g) $\{x \mid \|x\| \geq 1\}$
 - (h) $\{x \mid \|x\| < 1\}$
 - (i) $S^1 \cup (\mathbb{R}_+ \times 0)$
 - (j) $S^1 \cup (\mathbb{R}_+ \times \mathbb{R})$
 - (k) $S^1 \cup (\mathbb{R} \times 0)$
 - (l) $\mathbb{R}^2 - (\mathbb{R}_+ \times 0)$
3. Show that given a collection $\mathcal{C}$ of spaces, the relation of homotopy equivalence is an equivalence relation on $\mathcal{C}$.
 4. Let X be the figure eight and let Y be the theta space. Describe maps $f : X \rightarrow Y$ and $g : Y \rightarrow X$ that are homotopy inverse to each other.
 5. Recall that a space X is said to be *contractible* if the identity map of X to itself is nulhomotopic. Show that X is contractible if and only if X has the homotopy type of a one-point space.
 6. Show that a retract of a contractible space is contractible.
 7. Let A be a subspace of X ; let $j : A \rightarrow X$ be the inclusion map, and let $f : X \rightarrow A$ be a continuous map. Suppose there is a homotopy $H : X \times I \rightarrow X$ between the map $j \circ f$ and the identity map of X .
 - (a) Show that if f is a retraction, then j_* is an isomorphism.
 - (b) Show that if H maps $A \times I$ into A , then j_* is an isomorphism.
 - (c) Give an example in which j_* is not an isomorphism.
 - *8. Find a space X and a point x_0 of X such that inclusion $\{x_0\} \rightarrow X$ is a homotopy equivalence, but $\{x_0\}$ is not a deformation retract of X . [Hint: Let X be the subspace of $\mathbb{R}^2$ that is the union of the line segments $(1/n) \times I$, for $n \in \mathbb{Z}_+$, the line segment $0 \times I$, and the line segment $I \times 0$; let x_0 be the point $(0, 1)$. If $\{x_0\}$ is a deformation retract of X , show that for any neighborhood U of x_0 , the path component of U containing x_0 contains a neighborhood of x_0 .]
 9. We define the *degree* of a continuous map $h : S^1 \rightarrow S^1$ as follows:
 Let b_0 be the point $(1, 0)$ of S^1 ; choose a generator γ for the infinite cyclic group $\pi_1(S^1, b_0)$. If x_0 is any point of S^1 , choose a path α in S^1 from b_0 to x_0 ,

and define $\gamma(x_0) = \hat{\alpha}(\gamma)$. Then $\gamma(x_0)$ generates $\pi_1(S^1, x_0)$. The element $\gamma(x_0)$ is independent of the choice of the path α , since the fundamental group of S^1 is abelian.

Now given $h : S^1 \rightarrow S^1$, choose $x_0 \in S^1$ and let $h(x_0) = x_1$. Consider the homomorphism

$$h_* : \pi_1(S^1, x_0) \longrightarrow \pi_1(S^1, x_1).$$

Since both groups are infinite cyclic, we have

$$(*) \quad h_*(\gamma(x_0)) = d \cdot \gamma(x_1)$$

for some integer d , if the group is written additively. The integer d is called the **degree** of h and is denoted by $\deg h$.

The degree of h is independent of the choice of the generator γ ; choosing the other generator would merely change the sign of both sides of (*).

- (a) Show that d is independent of the choice of x_0 .
- (b) Show that if $h, k : S^1 \rightarrow S^1$ are homotopic, they have the same degree.
- (c) Show that $\deg(h \circ k) = (\deg h) \cdot (\deg k)$.
- (d) Compute the degrees of the constant map, the identity map, the reflection map $\rho(x_1, x_2) = (x_1, -x_2)$, and the map $h(z) = z^n$, where z is a complex number.
- *(e) Show that if $h, k : S^1 \rightarrow S^1$ have the same degree, they are homotopic.

10. Suppose that to every map $h : S^n \rightarrow S^n$ we have assigned an integer, denoted by $\deg h$ and called the **degree** of h , such that:

- (i) Homotopic maps have the same degree.
- (ii) $\deg(h \circ k) = (\deg h) \cdot (\deg k)$.
- (iii) The identity map has degree 1, any constant map has degree 0, and the reflection map $\rho(x_1, \dots, x_{n+1}) = (x_1, \dots, x_n, -x_{n+1})$ has degree -1 .

[One can construct such a function, using the tools of algebraic topology. Intuitively, $\deg h$ measures how many times h wraps S^n about itself; the sign tells you whether h preserves orientation or not.] Prove the following:

- (a) There is no retraction $r : B^{n+1} \rightarrow S^n$.
- (b) If $h : S^n \rightarrow S^n$ has degree different from $(-1)^{n+1}$, then h has a fixed point. [Hint: Show that if h has no fixed point, then h is homotopic to the antipodal map $a(x) = -x$.]
- (c) If $h : S^n \rightarrow S^n$ has degree different from 1, then h maps some point x to its antipode $-x$.
- (d) If S^n has a nonvanishing tangent vector field v , then n is odd. [Hint: If v exists, show the identity map is homotopic to the antipodal map.]

§59 The Fundamental Group of S^n

Now we turn to a problem mentioned at the beginning of the chapter, the problem of showing that the sphere, torus, and double torus are surfaces that are topologically distinct. We begin with the sphere; we show that S^n is simply connected for $n \geq 2$. The crucial result we need is stated in the following theorem.

Theorem 59.1. Suppose $X = U \cup V$, where U and V are open sets of X . Suppose that $U \cap V$ is path connected, and that $x_0 \in U \cap V$. Let i and j be the inclusion mappings of U and V , respectively, into X . Then the images of the induced homomorphisms

$$i_* : \pi_1(U, x_0) \rightarrow \pi_1(X, x_0) \quad \text{and} \quad j_* : \pi_1(V, x_0) \rightarrow \pi_1(X, x_0)$$

generate $\pi_1(X, x_0)$.

Proof. This theorem states that, given any loop f in X based at x_0 , it is path homotopic to a product of the form $(g_1 * (g_2 * (\cdots * g_n)))$, where each g_i is a loop in X based at x_0 that lies either in U or in V .

Step 1. We show there is a subdivision $a_0 < a_1 < \cdots < a_n$ of the unit interval such that $f(a_i) \in U \cap V$ and $f([a_{i-1}, a_i])$ is contained either in U or in V , for each i .

To begin, choose a subdivision $b_0, \dots, b_m$ of $[0, 1]$ such that for each i , the set $f([b_{i-1}, b_i])$ is contained in either U or V . (Use the Lebesgue number lemma.) If $f(b_i)$ belongs to $U \cap V$ for each i , we are finished. If not, let i be an index such that $f(b_i) \notin U \cap V$. Each of the sets $f([b_{i-1}, b_i])$ and $f([b_i, b_{i+1}])$ lies either in U or in V . If $f(b_i) \in U$, then both of these sets must lie in U ; and if $f(b_i) \in V$, both of them must lie in V . In either case, we may delete b_i , obtaining a new subdivision $c_0, \dots, c_{m-1}$ that still satisfies the condition that $f([c_{i-1}, c_i])$ is contained either in U or in V , for each i .

A finite number of repetitions of this process leads to the desired subdivision.

Step 2. We prove the theorem. Given f , let $a_0, \dots, a_n$ be the subdivision constructed in Step 1. Define f_i to be the path in X that equals the positive linear map of $[0, 1]$ onto $[a_{i-1}, a_i]$ followed by f . Then f_i is a path that lies either in U or in V , and by Theorem 51.3,

$$[f] = [f_1] * [f_2] * \cdots * [f_n].$$

For each i , choose a path α_i in $U \cap V$ from x_0 to $f(a_i)$. (Here we use the fact that $U \cap V$ is path connected.) Since $f(a_0) = f(a_n) = x_0$, we can choose α_0 and α_n to be the constant path at x_0 . See Figure 59.1.

Now we set

$$g_i = (\alpha_{i-1} * f_i) * \bar{\alpha}_i$$

for each i . Then g_i is a loop in X based at x_0 whose image lies either in U or in V . Direct computation shows that

$$[g_1] * [g_2] * \cdots * [g_n] = [f_1] * [f_2] * \cdots * [f_n]. \quad \blacksquare$$

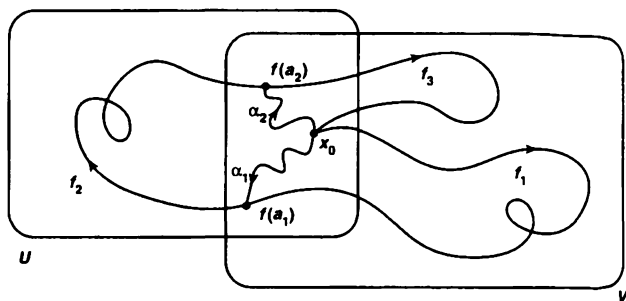


Figure 59.1

The preceding theorem is a special case of a famous theorem of topology called the *Seifert-van Kampen theorem*, which expresses the fundamental group of the space $X = U \cup V$ quite generally, when $U \cap V$ is path connected, in terms of the fundamental groups of U and V . We shall study this theorem in Chapter 11.

Corollary 59.2. Suppose $X = U \cup V$, where U and V are open sets of X ; suppose $U \cap V$ is nonempty and path connected. If U and V are simply connected, then X is simply connected.

Theorem 59.3. If $n \geq 2$, the n -sphere S^n is simply connected.

Proof. Let $p = (0, \dots, 0, 1) \in \mathbb{R}^{n+1}$ and $q = (0, \dots, 0, -1)$ be the “north pole” and the “south pole” of S^n , respectively.

Step 1. We show that if $n \geq 1$, the punctured sphere $S^n - p$ is homeomorphic to $\mathbb{R}^n$.

Define $f : (S^n - p) \rightarrow \mathbb{R}^n$ by the equation

$$f(x) = f(x_1, \dots, x_{n+1}) = \frac{1}{1 - x_{n+1}}(x_1, \dots, x_n).$$

The map f is called *stereographic projection*. (If one takes the straight line in $\mathbb{R}^{n+1}$ passing through the north pole p and the point x of $S^n - p$, then this line intersects the n -plane $\mathbb{R}^n \times 0 \subset \mathbb{R}^{n+1}$ in the point $f(x) \times 0$.) One checks that f is a homeomorphism by showing that the map $g : \mathbb{R}^n \rightarrow (S^n - p)$ given by

$$g(y) = g(y_1, \dots, y_n) = (t(y) \cdot y_1, \dots, t(y) \cdot y_n, 1 - t(y)),$$

where $t(y) = 2/(1 + \|y\|^2)$, is a right and left inverse for f .

Note that the reflection map $(x_1, \dots, x_{n+1}) \rightarrow (x_1, \dots, x_n, -x_{n+1})$ defines a homeomorphism of $S^n - p$ with $S^n - q$, so the latter is also homeomorphic to $\mathbb{R}^n$.

Step 2. We prove the theorem. Let U and V be the open sets $U = S^n - p$ and $V = S^n - q$ of S^n .

Note first that for $n \geq 1$, the sphere S^n is path connected. This follows from the fact that U and V are path connected (being homeomorphic to $\mathbb{R}^n$) and have the point $(1, 0, \dots, 0)$ of S^n in common.

Now we show that for $n \geq 2$, the sphere S^n is simply connected. The spaces U and V are simply connected, being homeomorphic to $\mathbb{R}^n$. Their intersection equals $S^n - p - q$, which is homeomorphic under stereographic projection to $\mathbb{R}^n - 0$. The latter space is path connected, for every point of $\mathbb{R}^n - 0$ can be joined to a point of S^{n-1} by a straight-line path, and S^{n-1} is path connected if $n \geq 2$. Then the preceding corollary applies. ■

Exercises

1. Let X be the union of two copies of S^2 having a single point in common. What is the fundamental group of X ? Prove that your answer is correct. [Be careful! The union of two simply connected spaces having a point in common is not necessarily simply connected. See [S], p. 59.]
2. Criticize the following "proof" that S^2 is simply connected: Let f be a loop in S^2 based at x_0 . Choose a point p of S^2 not lying in the image of f . Since $S^2 - p$ is homeomorphic with $\mathbb{R}^2$, and $\mathbb{R}^2$ is simply connected, the loop f is path homotopic to the constant loop.
3. (a) Show that $\mathbb{R}^1$ and $\mathbb{R}^n$ are not homeomorphic if $n > 1$.
(b) Show that $\mathbb{R}^2$ and $\mathbb{R}^n$ are not homeomorphic if $n > 2$.
It is, in fact, true that $\mathbb{R}^m$ and $\mathbb{R}^n$ are not homeomorphic if $n \neq m$, but the proof requires more advanced tools of algebraic topology.
4. Assume the hypotheses of Theorem 59.1.
(a) What can you say about the fundamental group of X if j_* is the trivial homomorphism? If both i_* and j_* are trivial?
(b) Give an example where i_* and j_* are trivial but neither U nor V have trivial fundamental groups

§60 Fundamental Groups of Some Surfaces

Recall that a *surface* is a Hausdorff space with a countable basis, each point of which has a neighborhood that is homeomorphic with an open subset of $\mathbb{R}^2$. Surfaces are of interest in various parts of mathematics, including geometry, topology, and complex analysis. We consider here several surfaces, including the torus and double torus, and show by comparing their fundamental groups that they are not homeomorphic. In a later chapter, we shall classify up to homeomorphism all compact surfaces.

First, we consider the torus. In an earlier exercise, we asked you to compute its fundamental group using the theory of covering spaces. Here, we compute its fundamental group by using a theorem about the fundamental group of a product space.

Recall that if A and B are groups with operation $\cdot$, then the cartesian product $A \times B$ is given a group structure by using the operation

$$(a \times b) \cdot (a' \times b') = (a \cdot a') \times (b \cdot b').$$

Recall also that if $h : C \rightarrow A$ and $k : C \rightarrow B$ are group homomorphisms, then the map $\Phi : C \rightarrow A \times B$ defined by $\Phi(c) = h(c) \times k(c)$ is a group homomorphism.

Theorem 60.1. $\pi_1(X \times Y, x_0 \times y_0)$ is isomorphic with $\pi_1(X, x_0) \times \pi_1(Y, y_0)$.

Proof. Let $p : X \times Y \rightarrow X$ and $q : X \times Y \rightarrow Y$ be the projection mappings. If we use the base points indicated in the statement of the theorem, we have induced homomorphisms

$$p_* : \pi_1(X \times Y, x_0 \times y_0) \longrightarrow \pi_1(X, x_0),$$

$$q_* : \pi_1(X \times Y, x_0 \times y_0) \longrightarrow \pi_1(Y, y_0).$$

We define a homomorphism

$$\Phi : \pi_1(X \times Y, x_0 \times y_0) \longrightarrow \pi_1(X, x_0) \times \pi_1(Y, y_0)$$

by the equation

$$\Phi([f]) = p_*([f]) \times q_*([f]) = [p \circ f] \times [q \circ f].$$

We shall show that Φ is an isomorphism.

The map Φ is surjective. Let $g : I \rightarrow X$ be a loop based at x_0 ; let $h : I \rightarrow Y$ be a loop based at y_0 . We wish to show that the element $[g] \times [h]$ lies in the image of Φ . Define $f : I \rightarrow X \times Y$ by the equation

$$f(s) = g(s) \times h(s).$$

Then f is a loop in $X \times Y$ based at $x_0 \times y_0$, and

$$\Phi([f]) = [p \circ f] \times [q \circ f] = [g] \times [h],$$

as desired.

The kernel of Φ vanishes. Suppose that $f : I \rightarrow X \times Y$ is a loop in $X \times Y$ based at $x_0 \times y_0$ and $\Phi([f]) = [p \circ f] \times [q \circ f]$ is the identity element. This means that $p \circ f \simeq_p e_{x_0}$ and $q \circ f \simeq_p e_{y_0}$; let G and H be the respective path homotopies. Then the map $F : I \times I \rightarrow X \times Y$ defined by

$$F(s, t) = G(s, t) \times H(s, t)$$

is a path homotopy between f and the constant loop based at $x_0 \times y_0$. ■

Corollary 60.2. *The fundamental group of the torus $T = S^1 \times S^1$ is isomorphic to the group $\mathbb{Z} \times \mathbb{Z}$.*

Now we define a surface called the projective plane and compute its fundamental group.

Definition. The *projective plane* P^2 is the quotient space obtained from S^2 by identifying each point x of S^2 with its antipodal point $-x$.

The projective plane may not be a space that is familiar to you; it cannot be imbedded in $\mathbb{R}^3$ and is thus difficult to visualize. It is, however, the fundamental object of study in projective geometry, just as the euclidean plane $\mathbb{R}^2$ is in ordinary euclidean geometry. Topologists are primarily interested in it as an example of a surface.

Theorem 60.3. *The projective plane P^2 is a compact surface, and the quotient map $p : S^2 \rightarrow P^2$ is a covering map.*

Proof. First we show that p is an open map. Let U be open in S^2 . Now the antipodal map $a : S^2 \rightarrow S^2$ given by $a(x) = -x$ is a homeomorphism of S^2 ; hence $a(U)$ is open in S^2 . Since

$$p^{-1}(p(U)) = U \cup a(U),$$

this set also is open in S^2 . Therefore, by definition, $p(U)$ is open in P^2 . A similar proof shows that p is a closed map.

Now we show that p is a covering map. Given a point y of P^2 , choose $x \in p^{-1}(y)$. Then choose an ϵ -neighborhood U of x in S^2 for some $\epsilon < 1$, using the euclidean metric d of $\mathbb{R}^3$. Then U contains no pair $\{z, a(z)\}$ of antipodal points of S^2 , since $d(z, a(z)) = 2$. As a result, the map

$$p : U \longrightarrow p(U)$$

is bijective. Being continuous and open, it is a homeomorphism. Similarly,

$$p : a(U) \rightarrow p(a(U)) = p(U)$$

is a homeomorphism. The set $p^{-1}(p(U))$ is thus the union of the two disjoint open sets U and $a(U)$, each of which is mapped homeomorphically by p onto $p(U)$. Then $p(U)$ is a neighborhood of $p(x) = y$ that is evenly covered by p .

Since S^2 has a countable basis $\{U_n\}$, the space P^2 has a countable basis $\{p(U_n)\}$.

The fact that P^2 is Hausdorff follows from the fact that S^2 is normal and p is a closed map. (See Exercise 6 of §31.) Alternatively, one can give a direct proof: Let y_1 and y_2 be two points of P^2 . The set $p^{-1}(y_1) \cup p^{-1}(y_2)$ consists of four points; let 2ϵ be the minimum distance between them. Let U_1 be the ϵ -neighborhood of one of the points of $p^{-1}(y_1)$, and let U_2 be the ϵ -neighborhood of one of the points of $p^{-1}(y_2)$. Then

$$U_1 \cup a(U_1) \quad \text{and} \quad U_2 \cup a(U_2)$$

are disjoint. It follows that $p(U_1)$ and $p(U_2)$ are disjoint neighborhoods of y_1 and y_2 , respectively, in P^2 .

Since S^2 is a surface and every point of P^2 has a neighborhood homeomorphic with an open subset of S^2 , the space P^2 is also a surface. ■

Corollary 60.4. $\pi_1(P^2, y)$ is a group of order 2.

Proof. The projection $p : S^2 \rightarrow P^2$ is a covering map. Since S^2 is simply connected, we can apply Theorem 54.4, which tells us there is a bijective correspondence between $\pi_1(P^2, y)$ and the set $p^{-1}(y)$. Since this set is a two-element set, $\pi_1(P^2, y)$ is a group of order 2.

Any group of order 2 is isomorphic to $\mathbb{Z}/2$, the integers mod 2, of course. ■

One can proceed similarly to define P^n , for any $n \in \mathbb{Z}_+$, as the space obtained from S^n by identifying each point x with its antipode $-x$; it is called **projective n -space**. The proof of Theorem 60.3 goes through without change to prove that the projection $p : S^n \rightarrow P^n$ is a covering map. Then because S^n is simply connected for $n \geq 2$, it follows that $\pi_1(P^n, y)$ is a two-element group for $n \geq 2$. We leave it to you to figure out what happens when $n = 1$.

Now we study the double torus. We begin with a lemma about the figure eight.

Lemma 60.5. *The fundamental group of the figure eight is not abelian.*

Proof. Let X be the union of two circles A and B in $\mathbb{R}^2$ whose intersection consists of the single point x_0 . We describe a certain covering space E of X .

The space E is the subspace of the plane consisting of the x -axis and the y -axis, along with tiny circles tangent to these axes, one circle tangent to the x -axis at each nonzero integer point and one circle tangent to the y -axis at each nonzero integer point.

The projection map $p : E \rightarrow X$ wraps the x -axis around the circle A and wraps the y -axis around the other circle B ; in each case the integer points are mapped by p into the base point x_0 . Each circle tangent to an integer point on the x -axis is mapped homeomorphically by p onto B , while each circle tangent to an integer point on the y -axis is mapped homeomorphically onto A ; in each case the point of tangency is mapped onto the point x_0 . We leave it to you to check mentally that the map p is indeed a covering map.

We could write this description down in equations if we wished, but the informal description seems to us easier to follow.

Now let $\tilde{f} : I \rightarrow E$ be the path $\tilde{f}(s) = s \times 0$, going along the x -axis from the origin to the point 1×0 . Let $\tilde{g} : I \rightarrow E$ be the path $\tilde{g}(s) = 0 \times s$, going along the y -axis from the origin to the point 0×1 . Let $f = p \circ \tilde{f}$ and $g = p \circ \tilde{g}$; then f and g are loops in the figure eight based at x_0 , going around the circles A and B , respectively. See Figure 60.1.

We assert that $f * g$ and $g * f$ are not path homotopic, so that the fundamental group of the figure eight is not abelian.

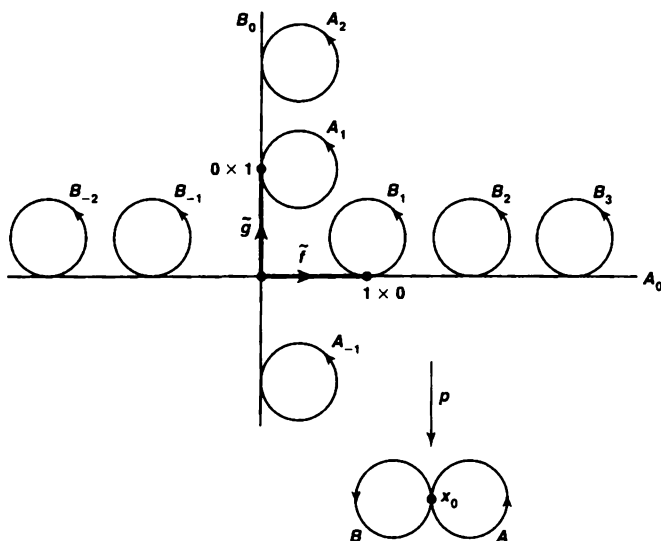


Figure 60.1

To prove this assertion, let us lift each of these to a path in E beginning at the origin. The path $f * g$ lifts to a path that goes along the x -axis from the origin to 1×0 and then goes once around the circle tangent to the x -axis at 1×0 . On the other hand, the path $g * f$ lifts to a path in E that goes along the y -axis from the origin to 0×1 , and then goes once around the circle tangent to the y -axis at 0×1 . Since the lifted paths do not end at the same point, $f * g$ and $g * f$ cannot be path homotopic. ■

We shall prove later that the fundamental group of the figure eight is, in fact, the group that algebraists call the “free group on two generators.”

Theorem 60.6. *The fundamental group of the double torus is not abelian.*

Proof. The double torus $T \# T$ is the surface obtained by taking two copies of the torus, deleting a small open disc from each of them, and pasting the remaining pieces together along their edges. We assert that the figure eight X is a retract of $T \# T$. This fact implies that inclusion $j : X \rightarrow T \# T$ induces a monomorphism j_* , so that $\pi_1(T \# T, x_0)$ is not abelian.

One can write equations for the retraction $r : T \# T \rightarrow X$, but it is simpler to indicate it in pictures, as we have done in Figure 60.2. Let Y be the union of two tori having a point in common. First one maps $T \# T$ onto Y by a map that collapses the dotted circle to a point but is otherwise one-to-one; it defines a homeomorphism h of

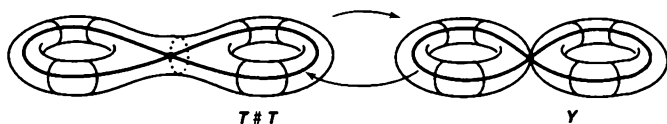


Figure 60.2

the figure eight in $T \# T$ with the figure eight in Y . Then one retracts Y onto its figure eight by mapping each cross-sectional circle to the point where it intersects the figure eight. Then one maps the figure eight in Y back onto the figure eight in $T \# T$ by the map h^{-1} . ■

Corollary 60.7. *The 2-sphere, torus, projective plane, and double torus are topologically distinct.*

Exercises

1. Compute the fundamental groups of the “solid torus” $S^1 \times B^2$ and the product space $S^1 \times S^2$.
2. Let X be the quotient space obtained from B^2 by identifying each point x of S^1 with its antipode $-x$. Show that X is homeomorphic to the projective plane P^2 .
3. Let $p : E \rightarrow X$ be the map constructed in the proof of Lemma 60.5. Let E' be the subspace of E that is the union of the x -axis and the y -axis. Show that $p|_{E'}$ is not a covering map.
4. The space P^1 and the covering map $p : S^1 \rightarrow P^1$ are familiar ones. What are they?
5. Consider the covering map indicated in Figure 60.3. Here, p wraps A_1 around A twice and wraps B_1 around B twice; p maps A_0 and B_0 homeomorphically onto A and B , respectively. Use this covering space to show that the fundamental group of the figure eight is not abelian.

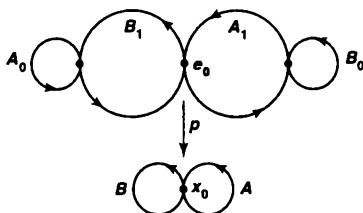


Figure 60.3